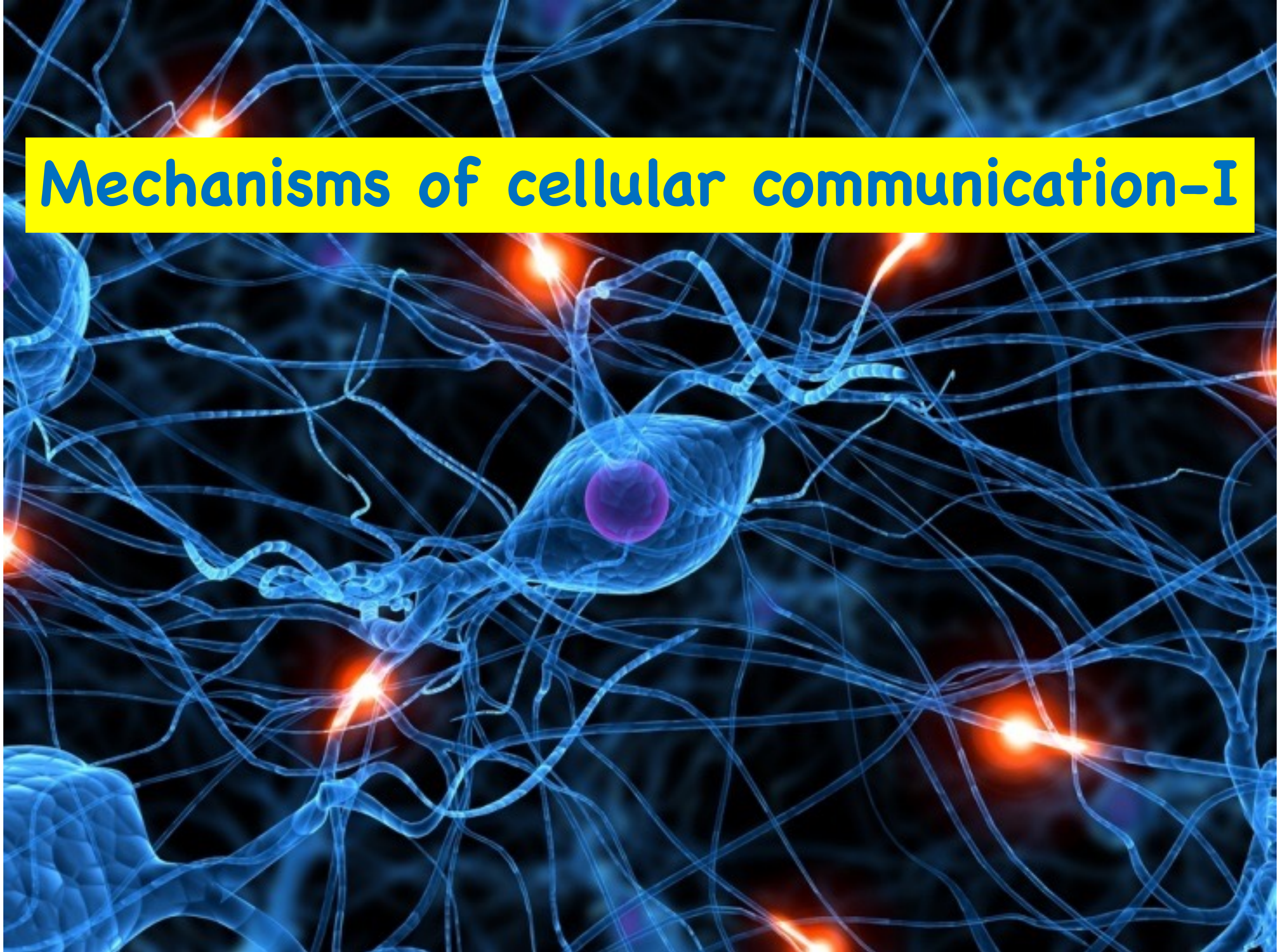
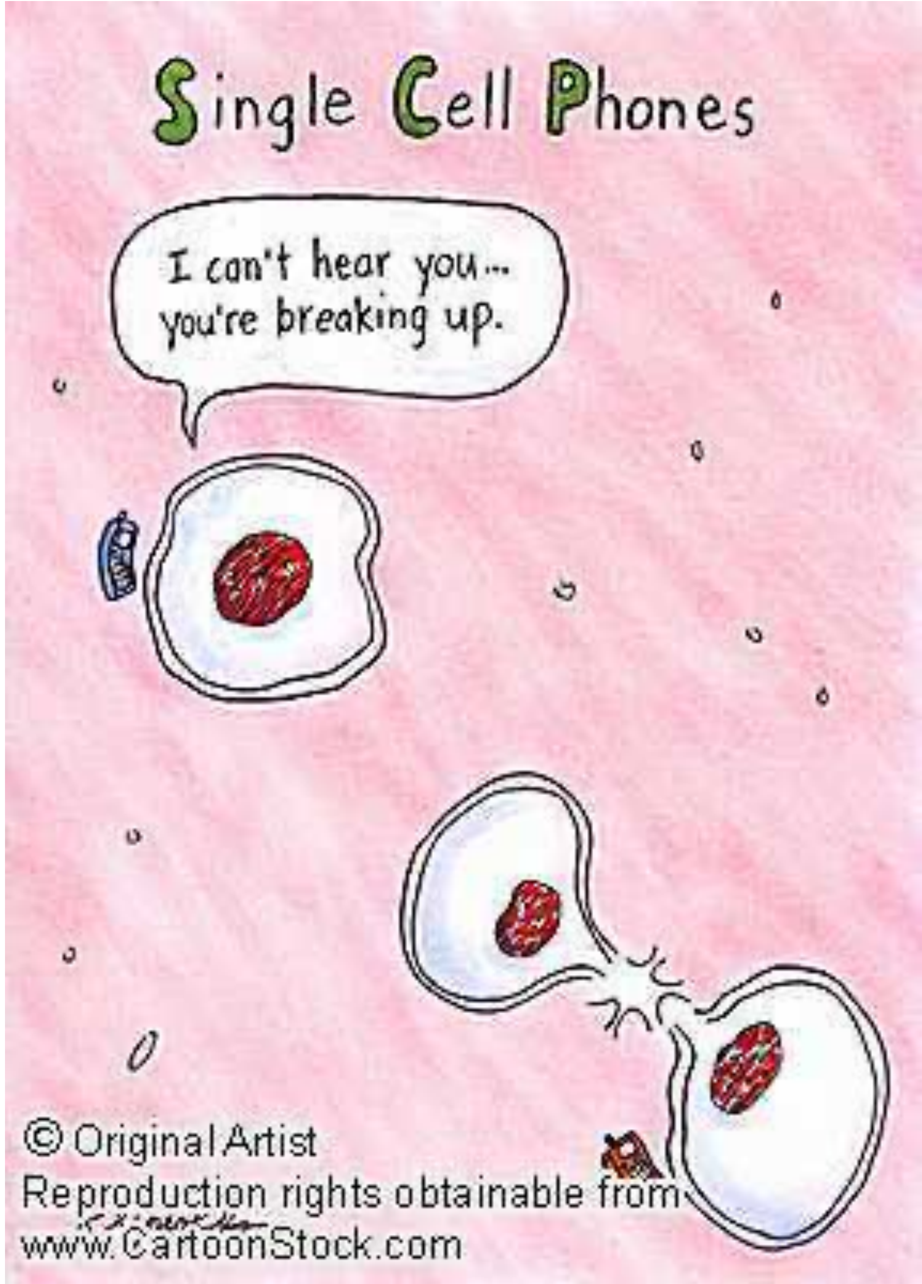
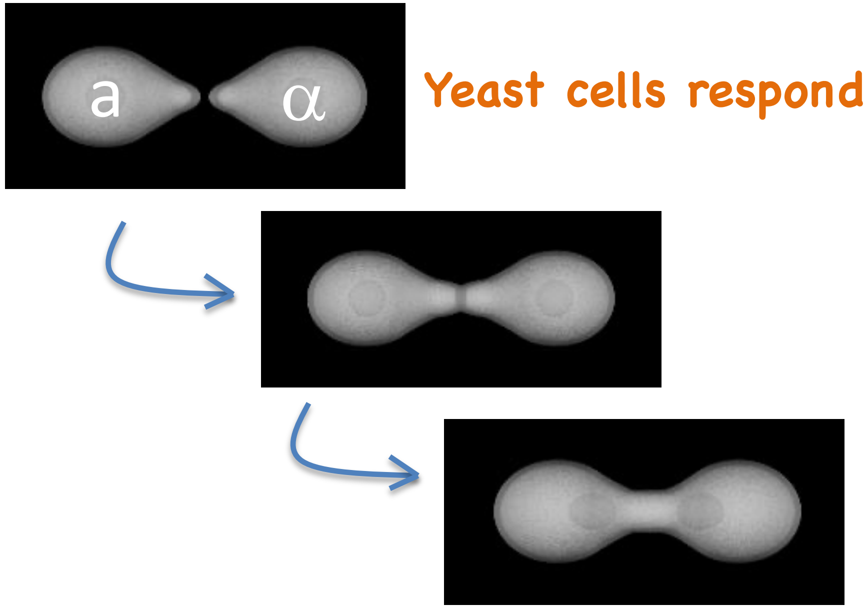
Mechanisms of cellular communication-I



Cell communication is very important



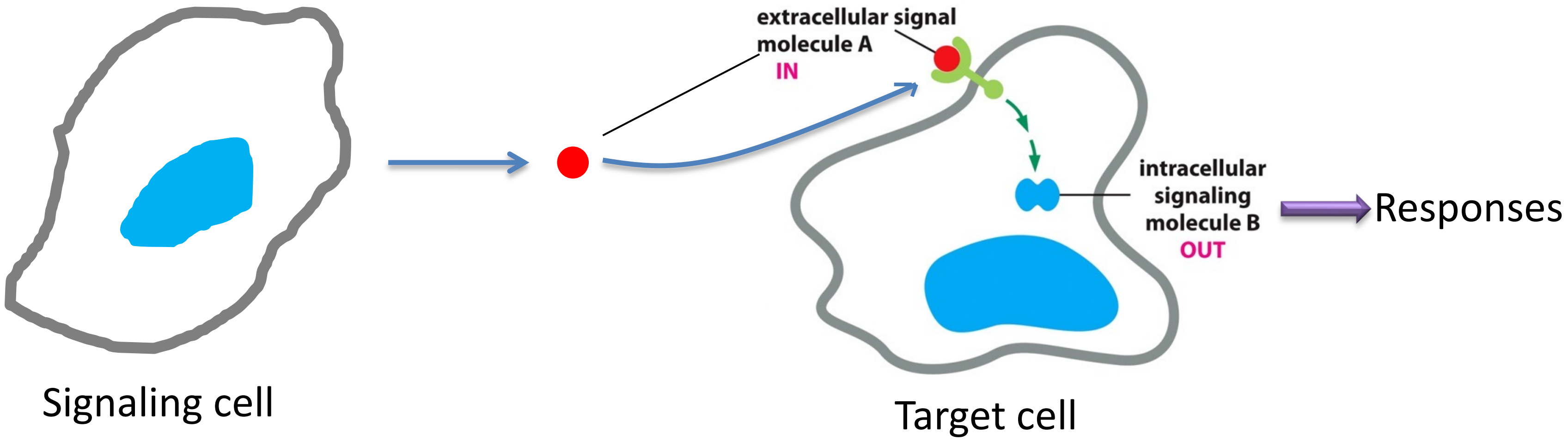
- Sense and respond to the environment
- Sending out orders
- Wound healing
- Development
- more and more….



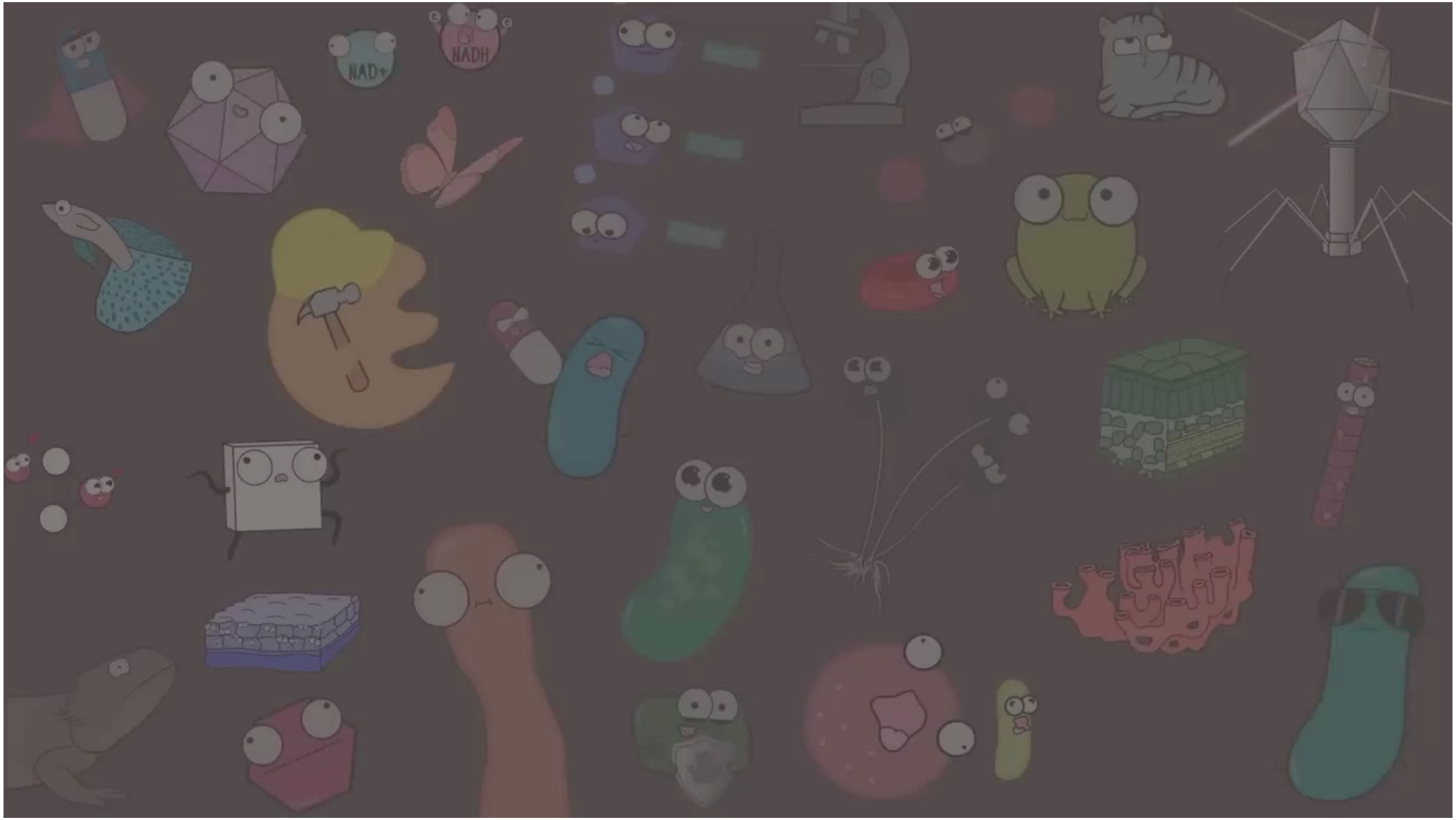
Yeast cells respond to mating factor

How cell communicate?

Signal transduction: external signal are converted to responses within the cells



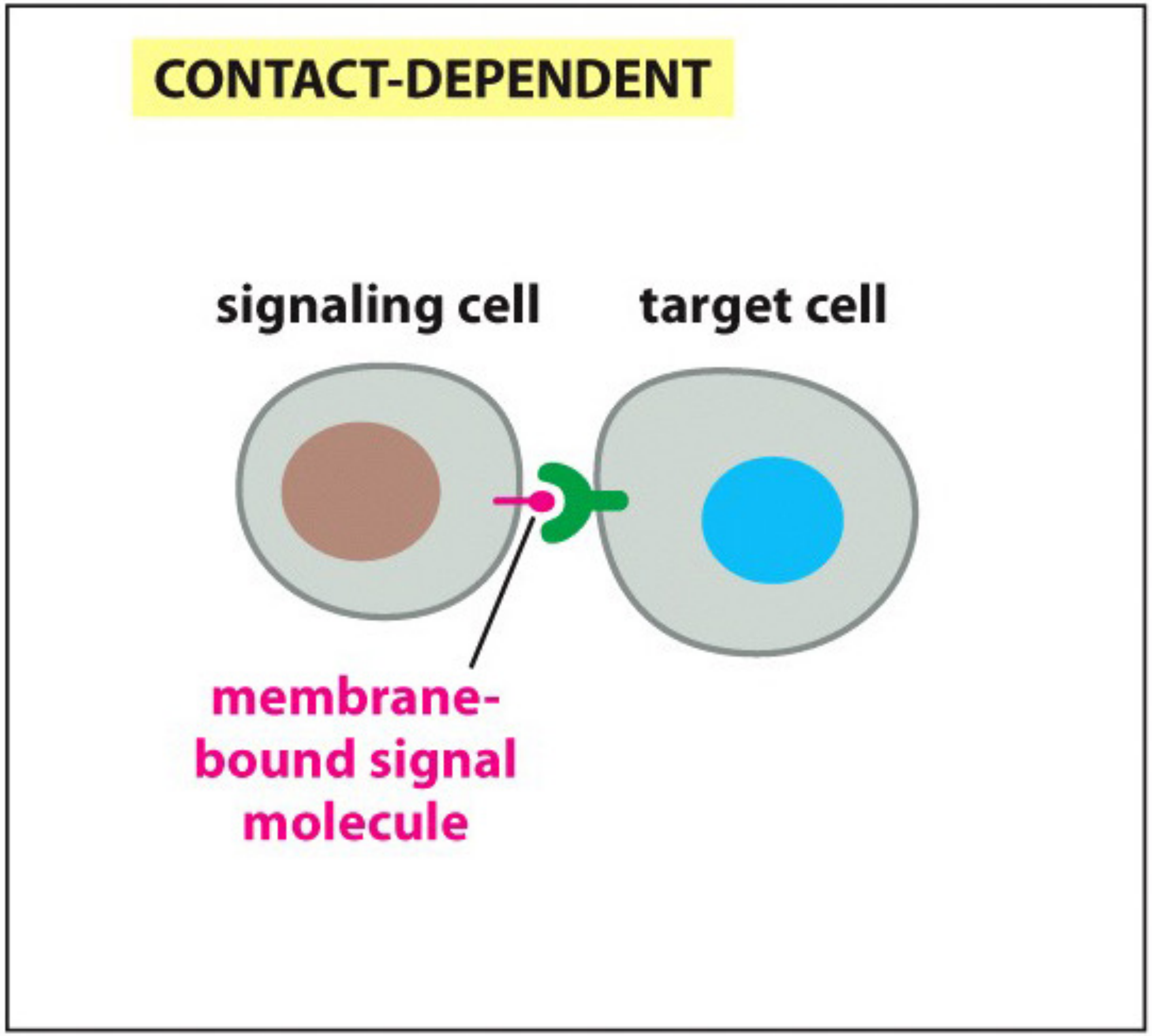
How cell communicate?

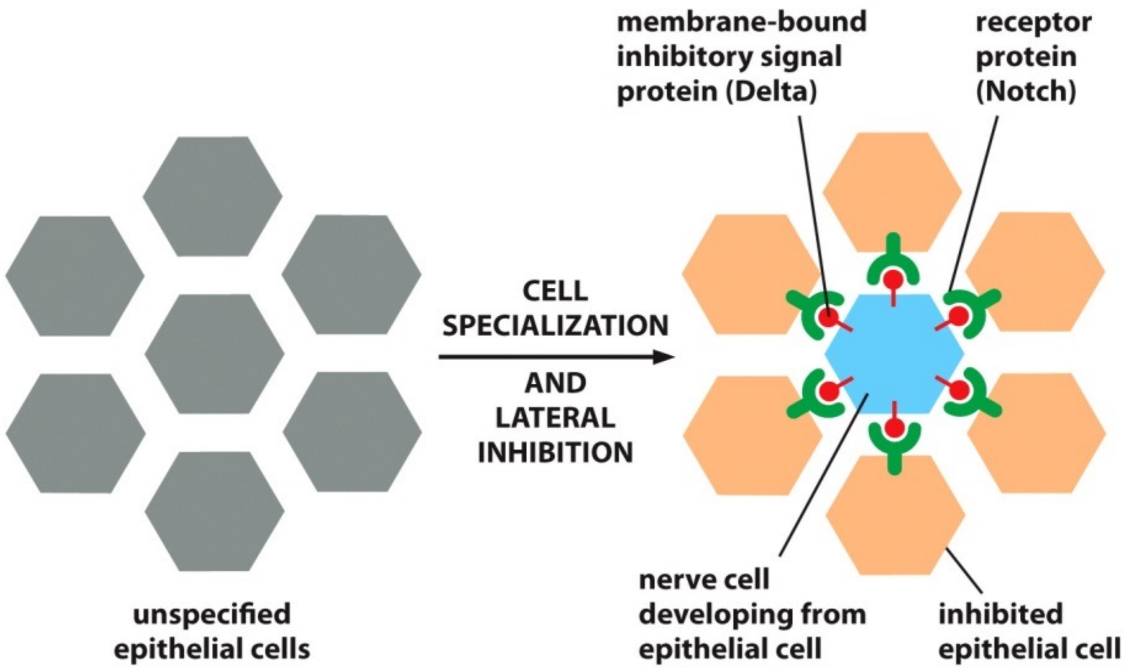


https://www.youtube.com/watch?v=-dbRterutHY

Cells communicate via various ways

Direct contact





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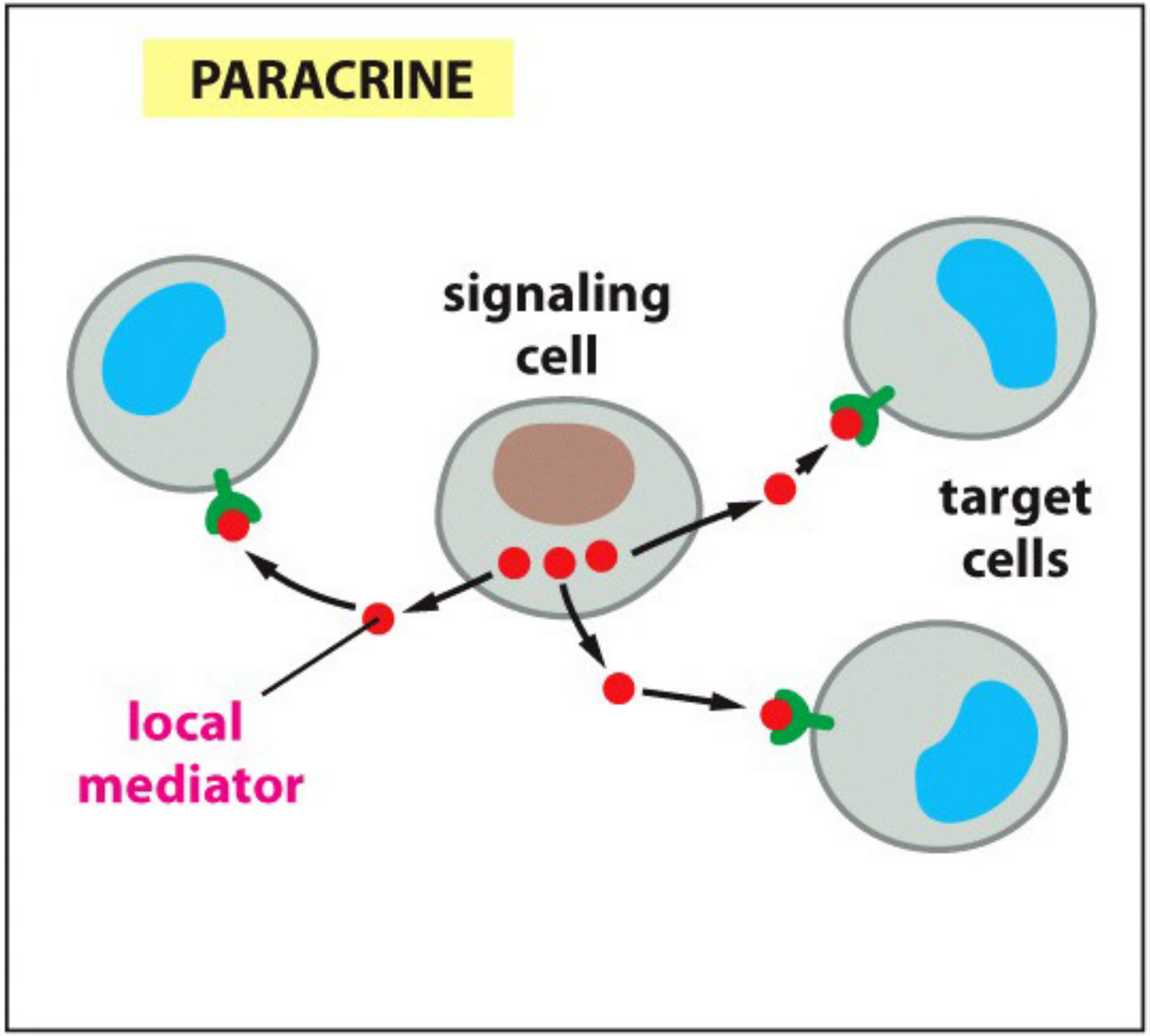
Allow adjacent cells to form different cell types

Cells communicate via various ways

Short distance

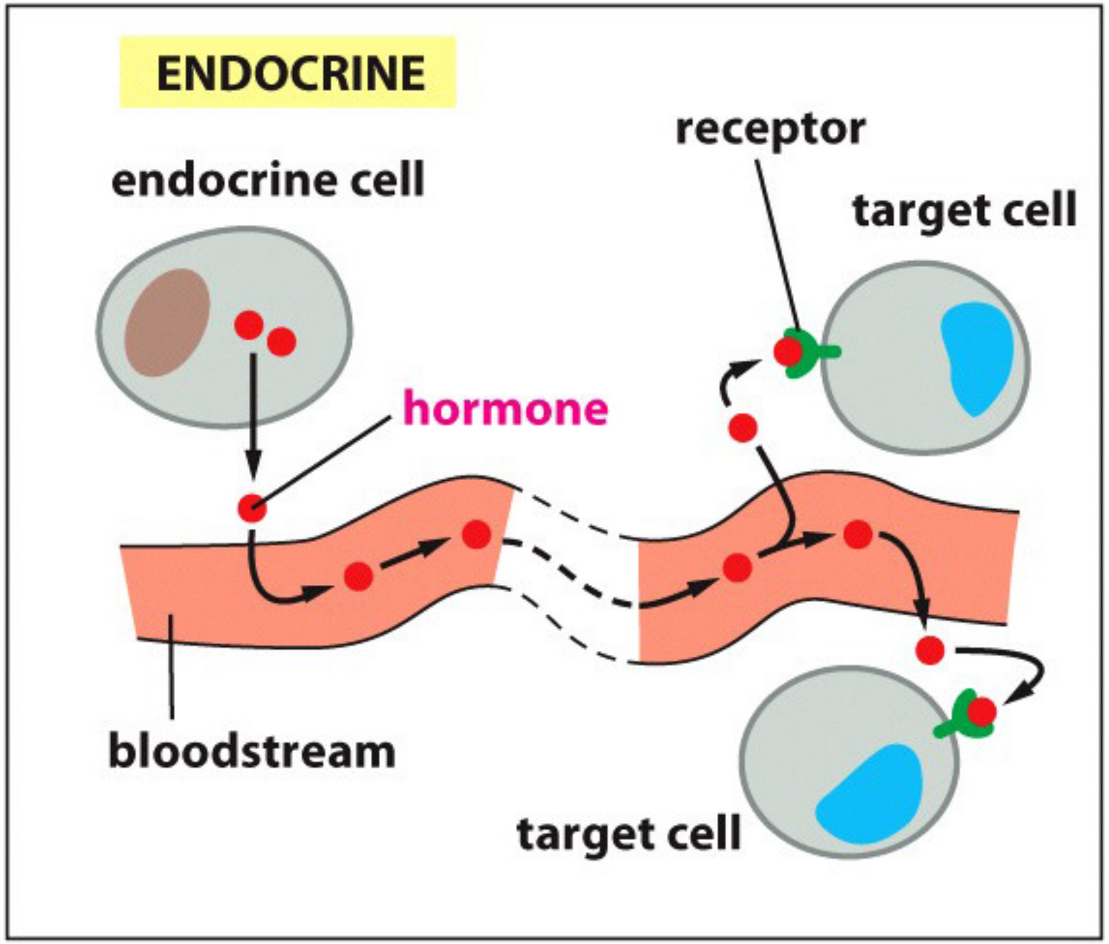


Description:
Signal molecules diffuse locally remaining in the neighborhood of the cell that secretes them.

Function (local mediators):
- Regulate inflammation at the infection area
- Control cell proliferation in a healing wound

Cells communicate via various ways

Long distance

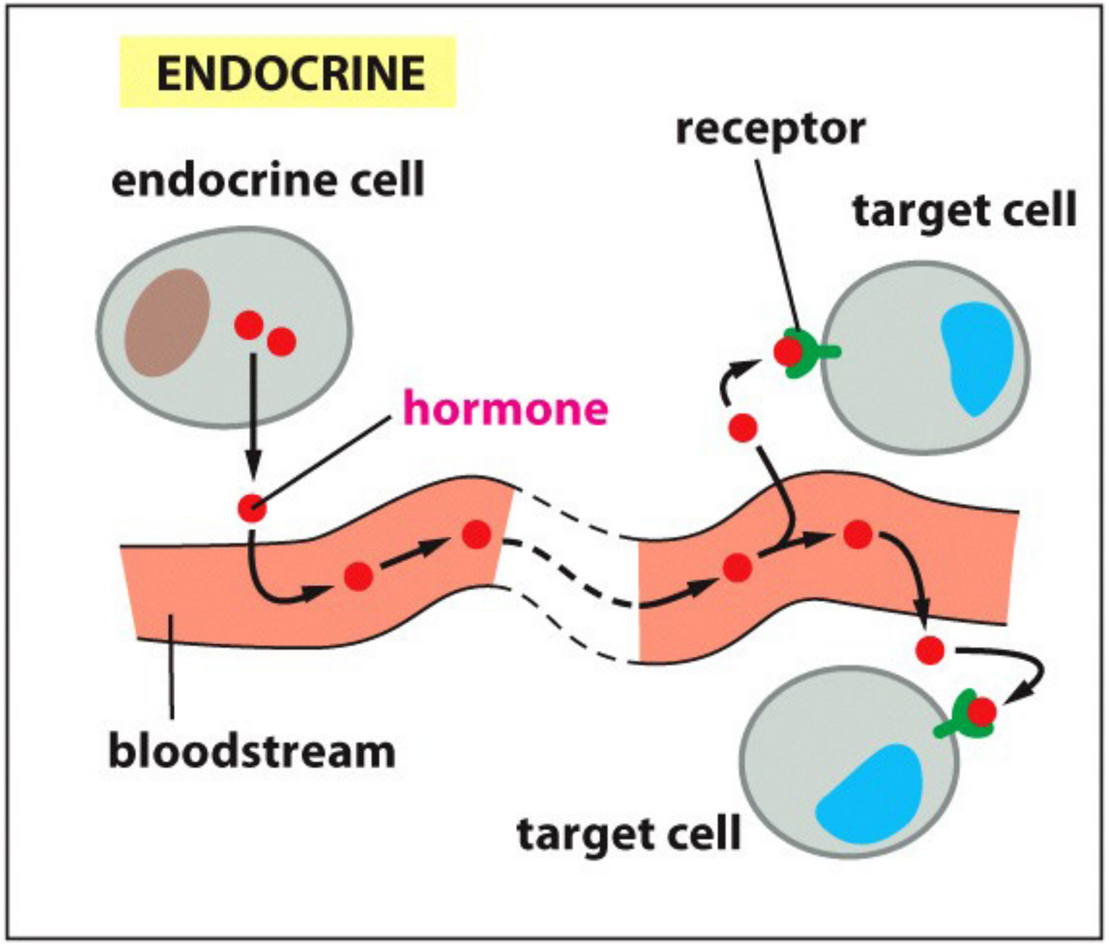


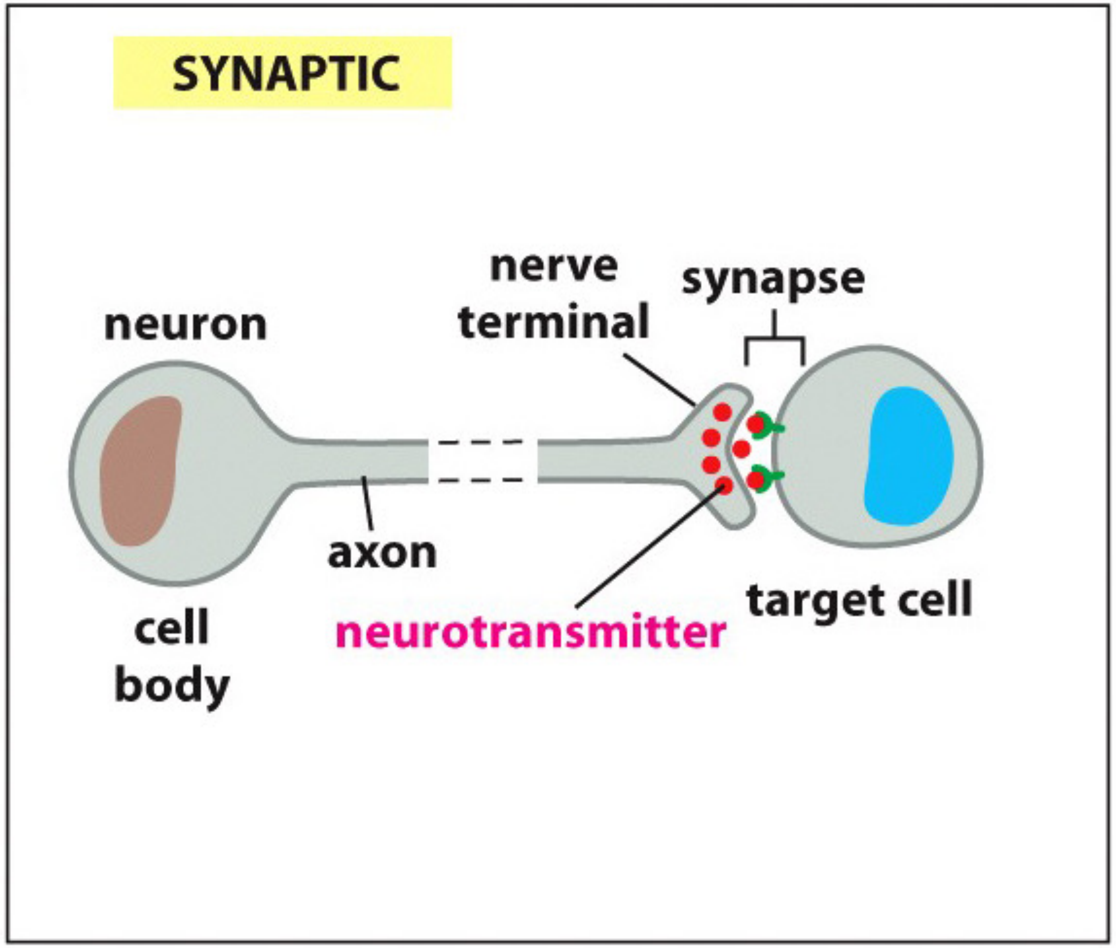
An example of signaling through hormones:

Pancreases produce and secrete insulin into the bloodstream that regulates glucose uptake in cells all over the body.

Cells communicate via various ways

Long distance





<u>An example of signaling through hormones:</u>

Pancreases produce and secrete insulin into the bloodstream that regulates glucose uptake in cells all over the body.

<u>Neuronal signaling:</u>

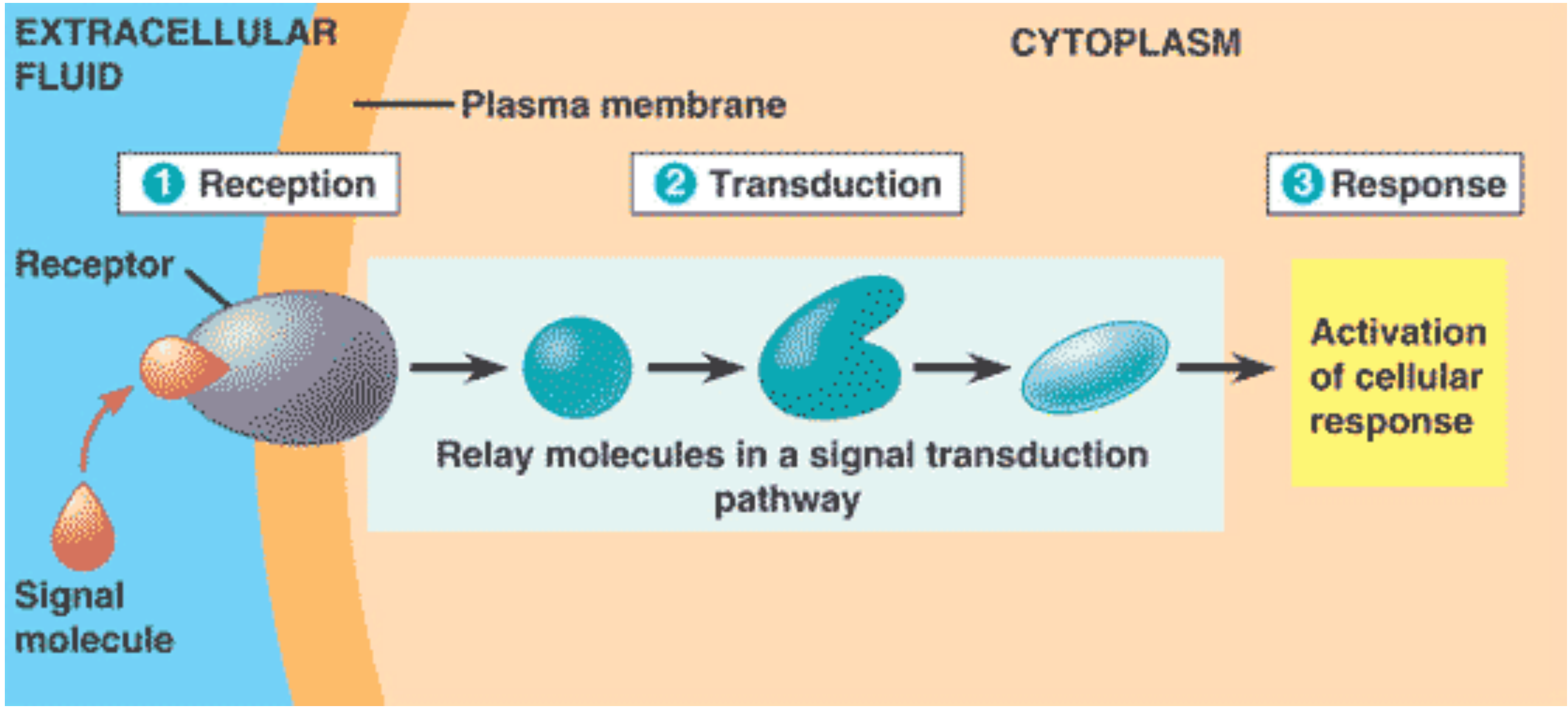
A messages (neurotransmitter) is delivered to the target cells quickly and specifically through private lines.

Three phases of signal transduction

1. Reception- the target cell detects the extracellular signal

2. Transduction- the conversion of the extracellular signal to a form that can bring a specific response

3. Response- the specific cellular response to the signal molecule



How extracellular signals are received?

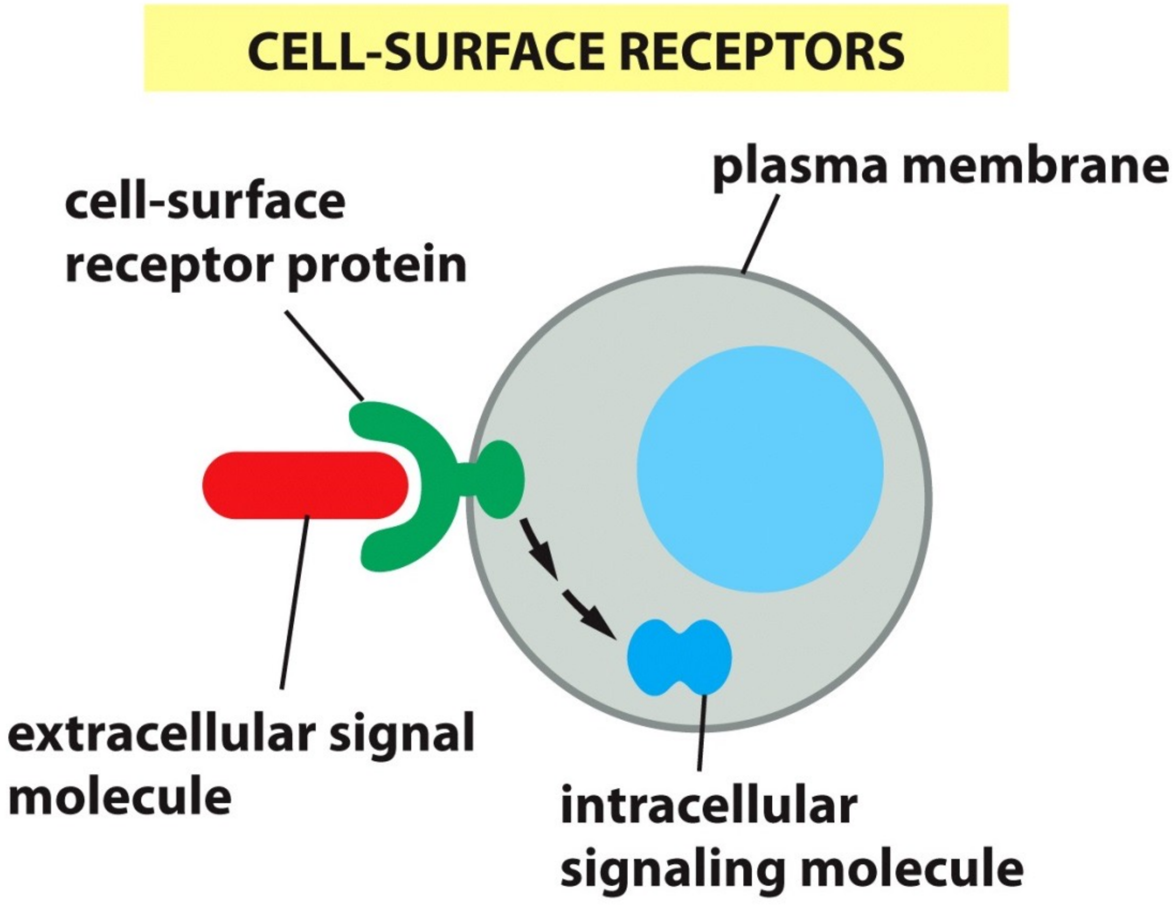


Figure 16-8a Essential Cell Biology, 4th ed. (© Garland Science 2014)

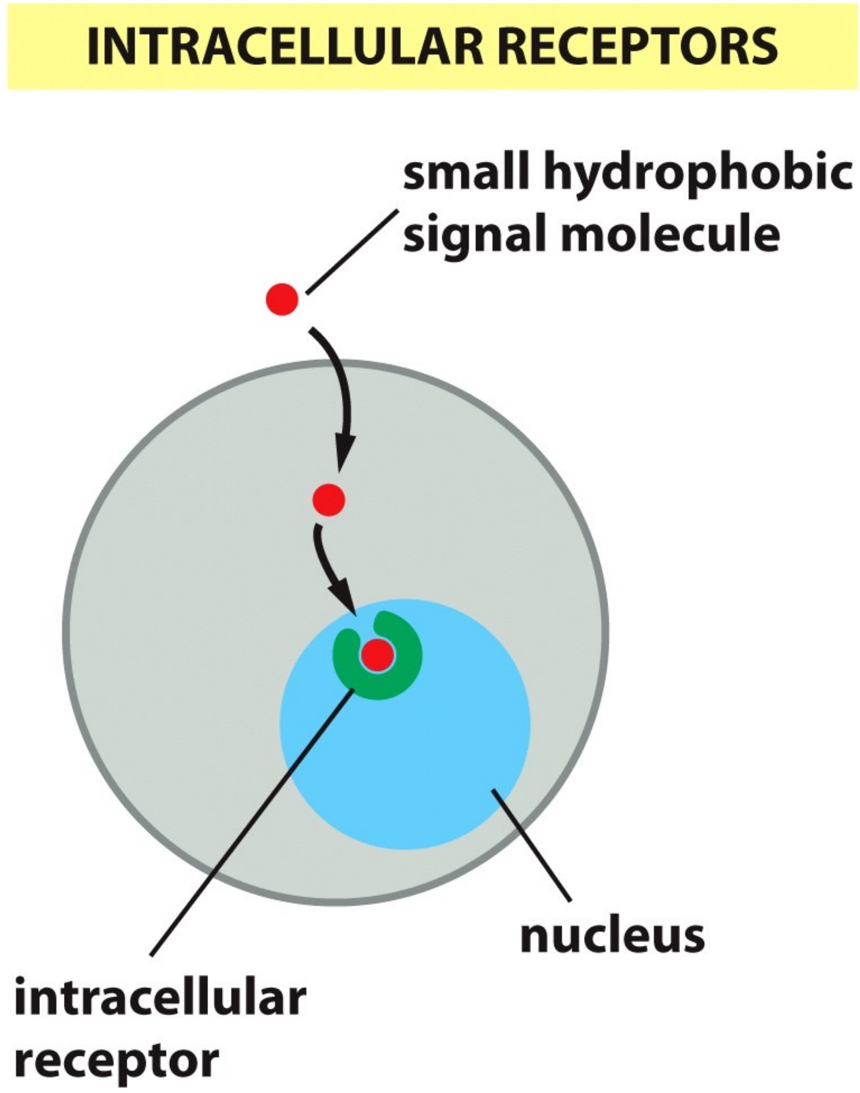


Figure 16-8b Essential Cell Biology, 4th ed. (© Garland Science 2014)

Extracellular signal molecules bind either to cell-surface receptors or to intracellular receptors.

Three phases of signal transduction

1. Reception- the target cell detects the extracellular signal

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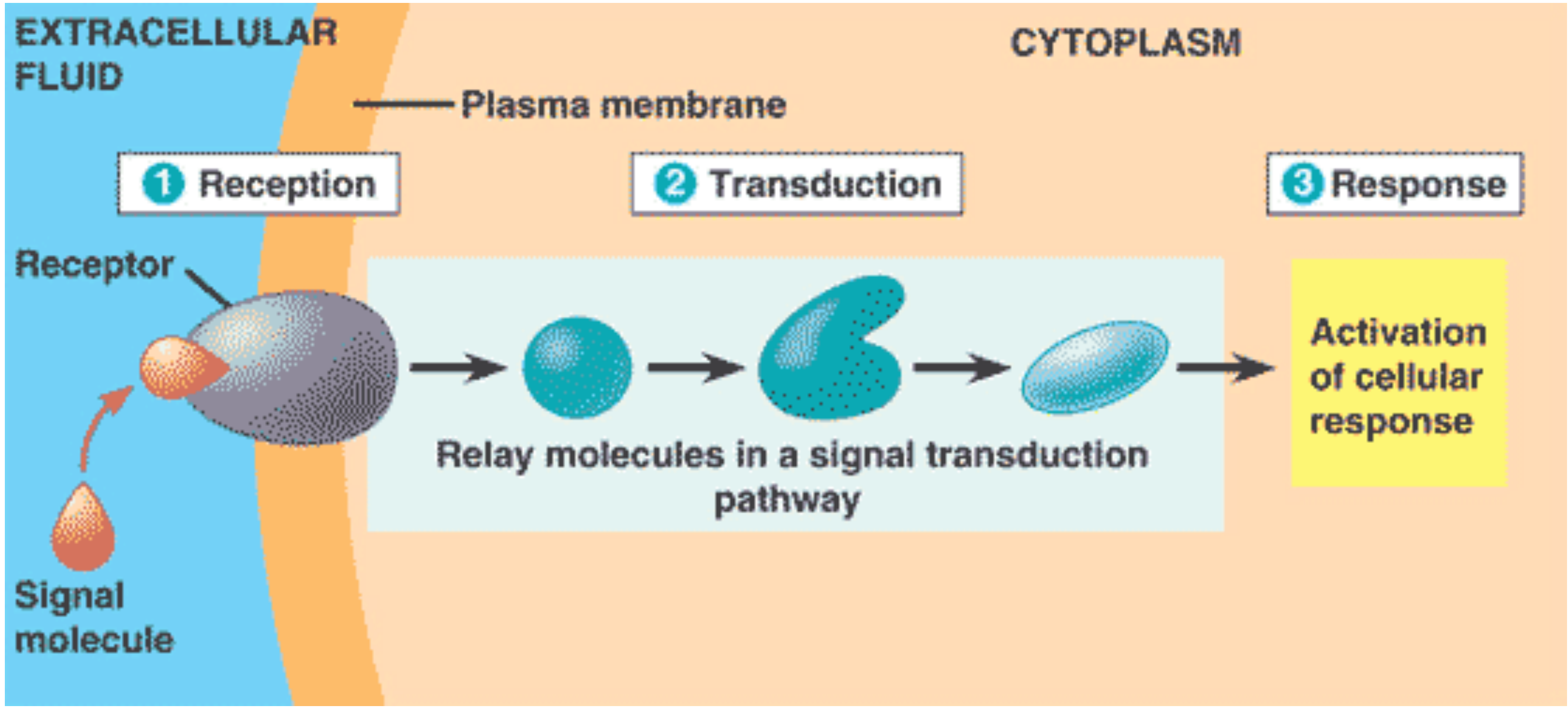


Extracellular signals act via cell surface receptors to induce a molecular relay leading to changes of behaviors of the target cells

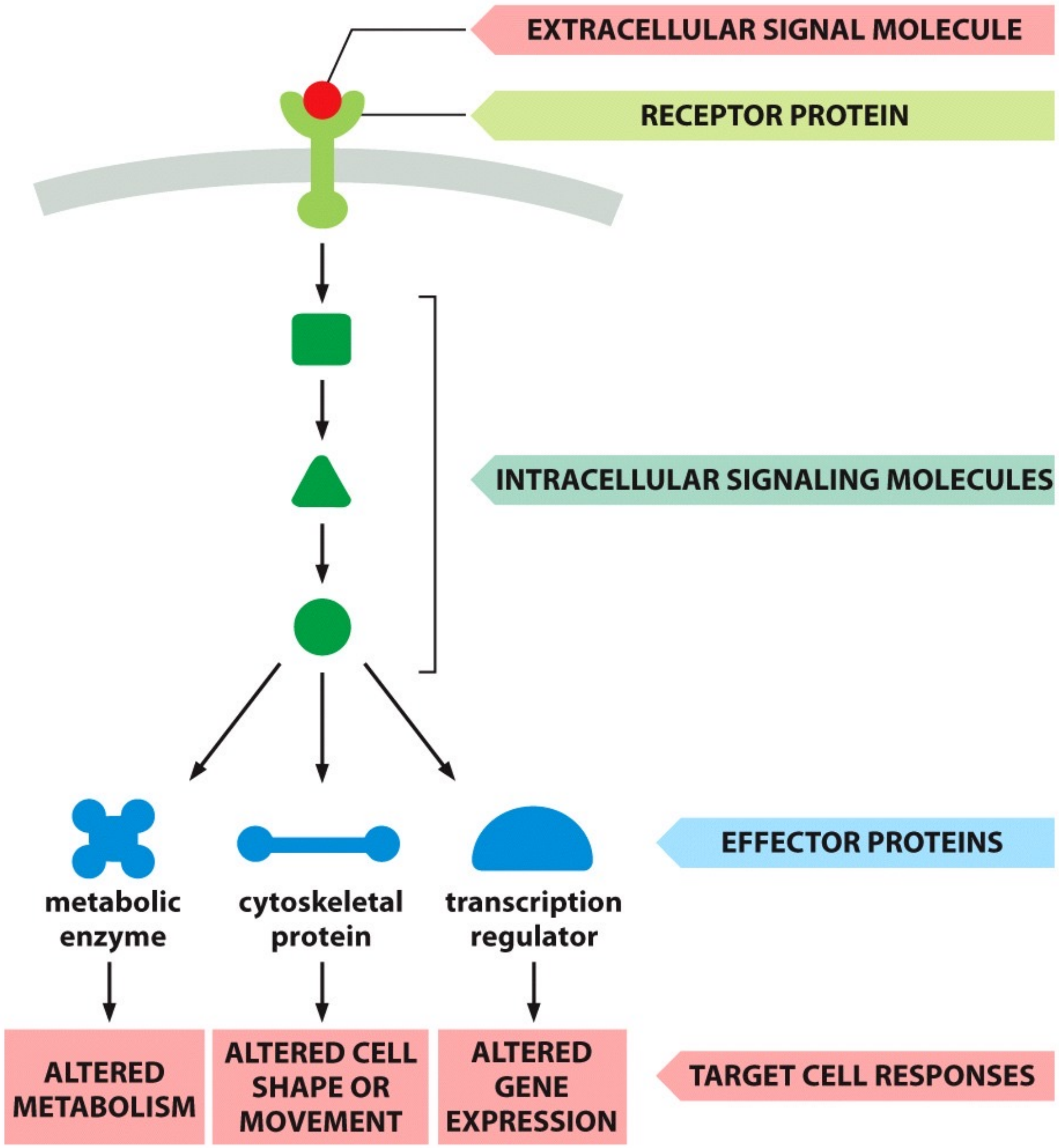
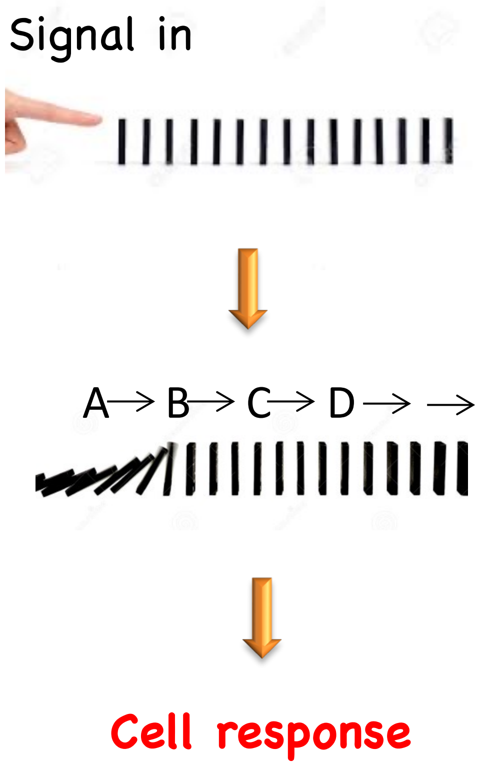
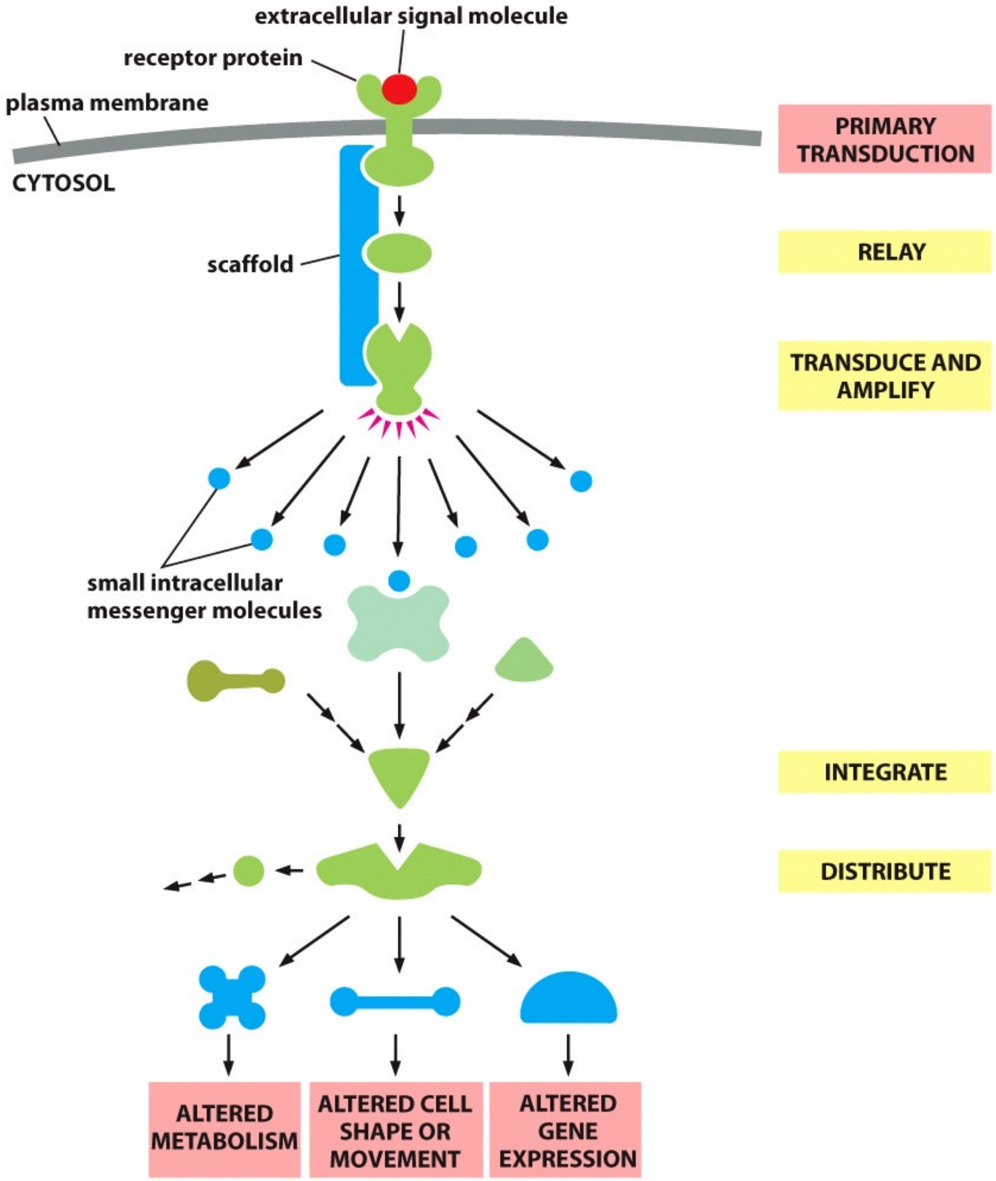


Figure 16-12 Essential Cell Biology, 4th ed. (© Garland Science 2014)

Domino Effect Chain Reaction



Intracellular signaling pathways respond to the extracellular signal in multiple ways



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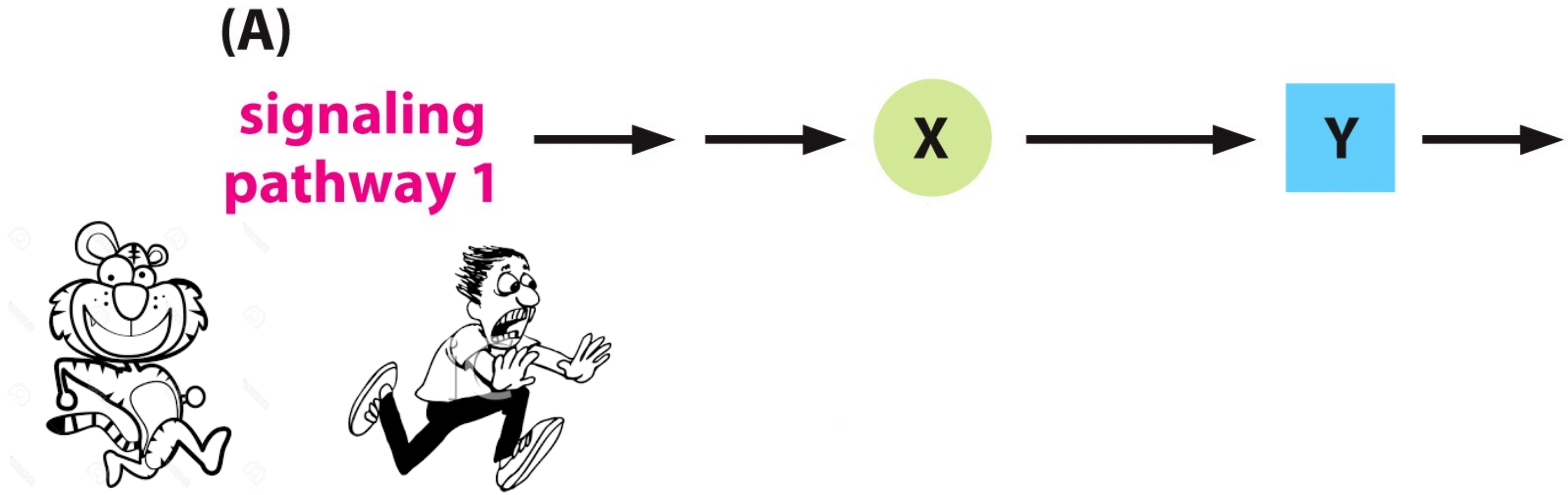
Signaling pathways are regulated by feedback regulation



Many intracellular signaling proteins act as molecular switches

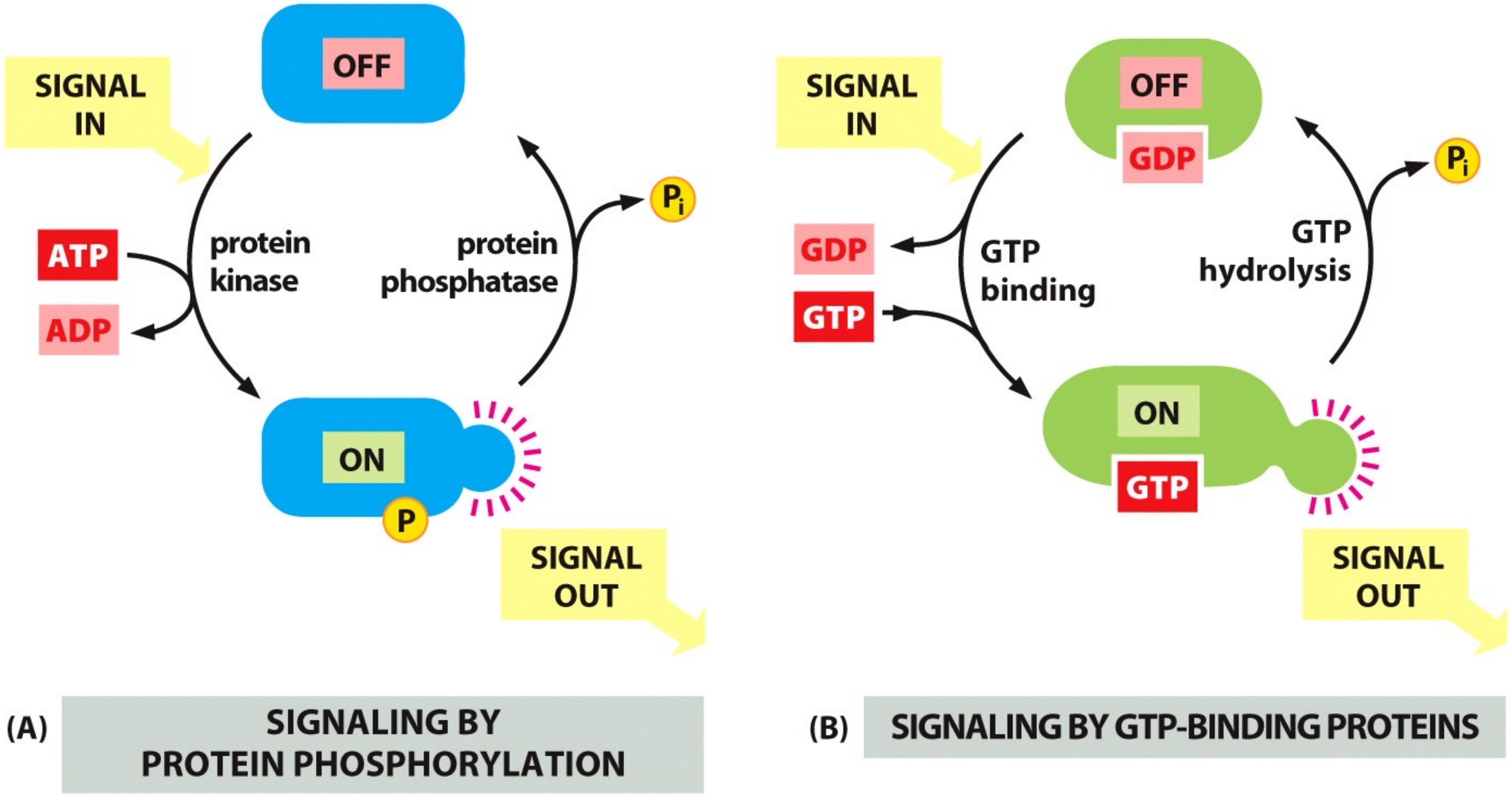


Figure 16-15 Essential Cell Biology, 4th ed. (© Garland Science 2014)

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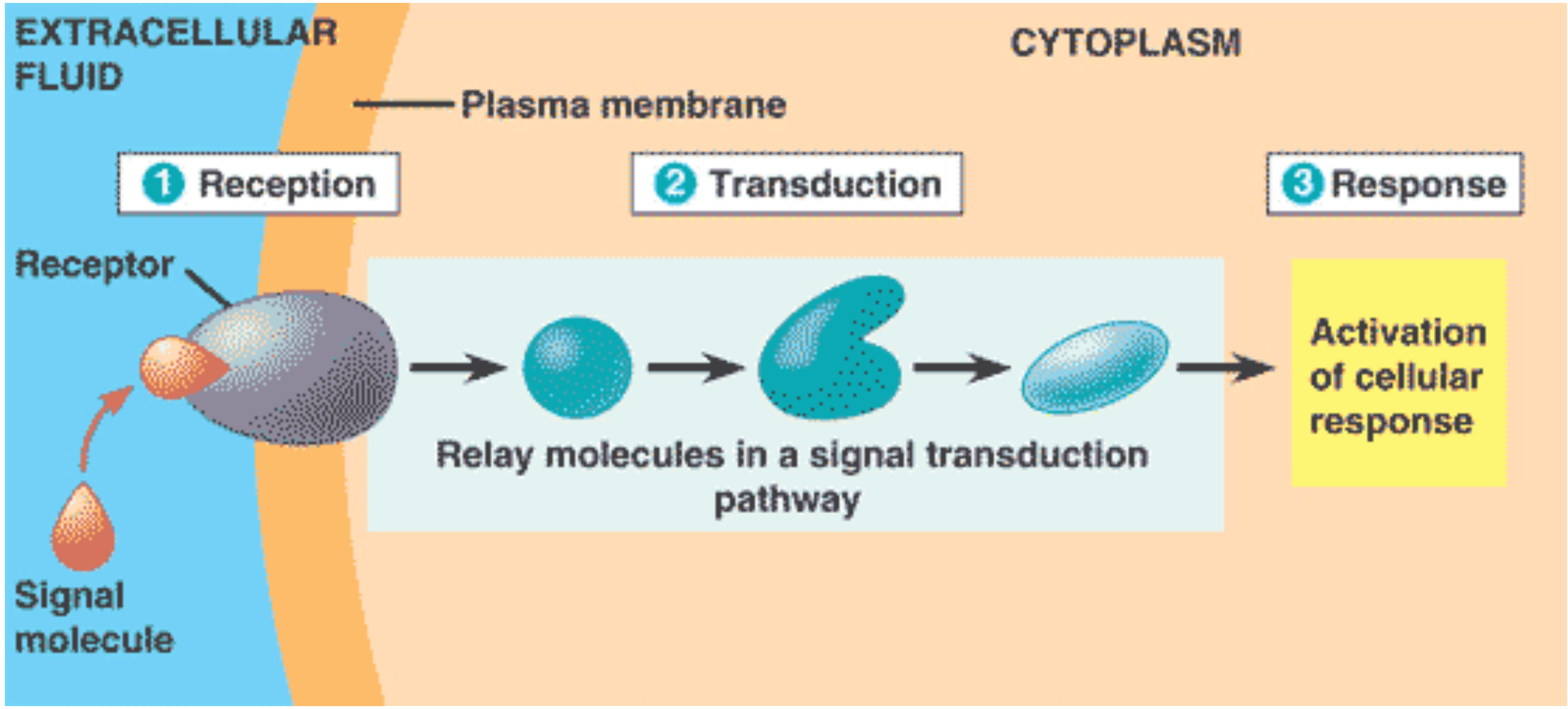
3. Response- the specific cellular response to the signal molecule



The same signal molecules can induce different responses in different cells

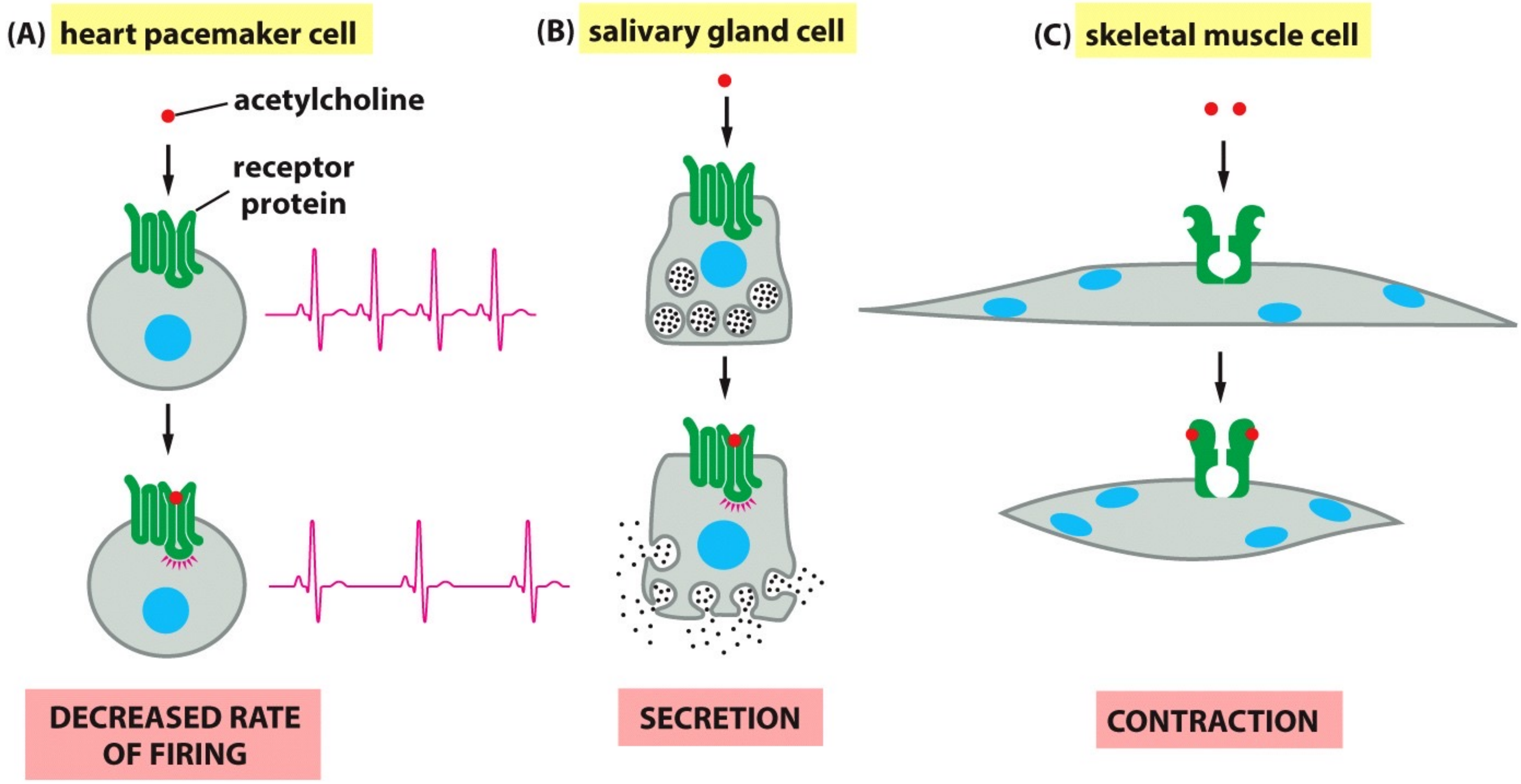


Figure 16-5abc Essential Cell Biology, 4th ed. (© Garland Science 2014)

Different combination of extracellular signals causes different responses of target cells

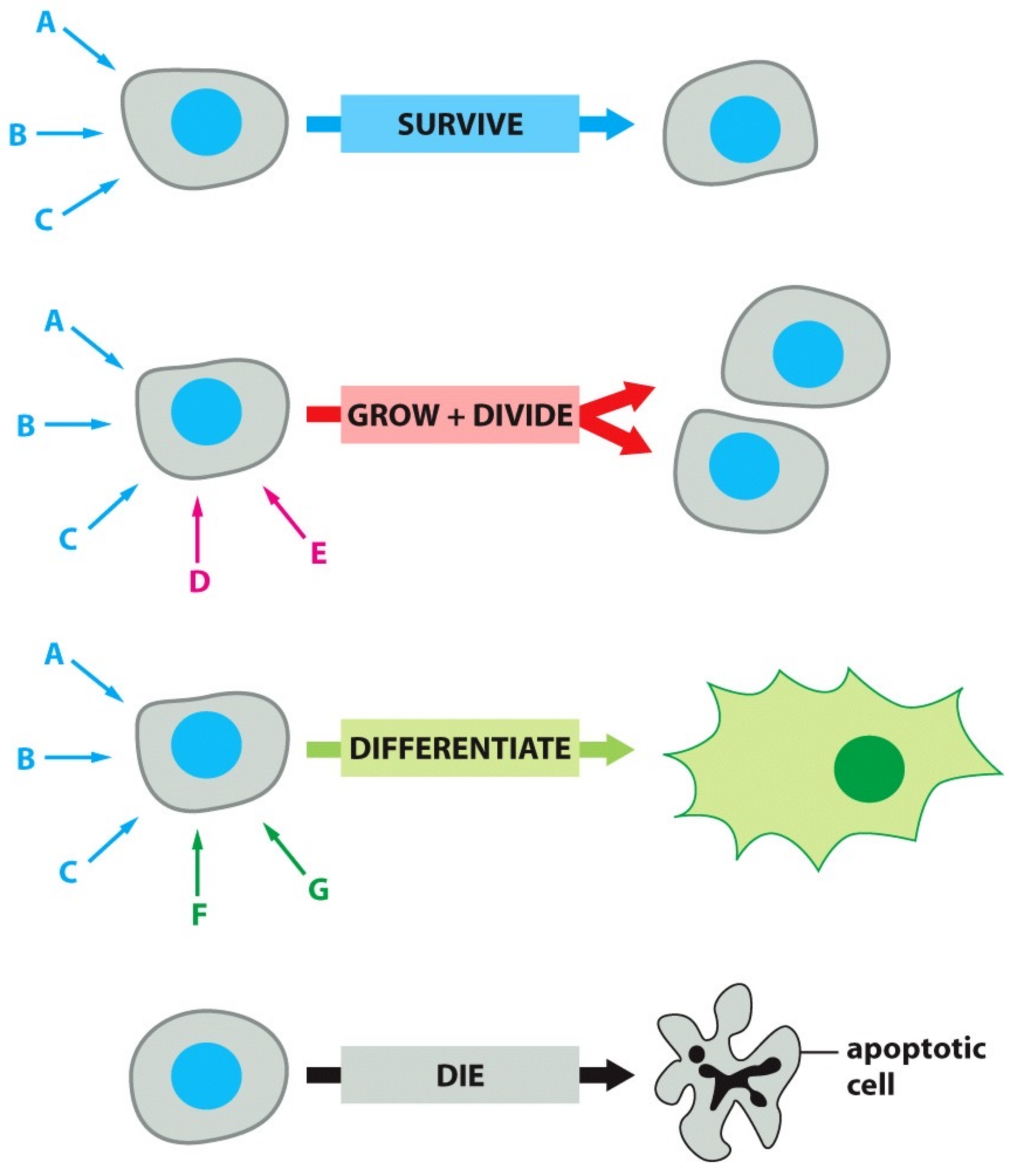


Figure 16-6 Essential Cell Biology, 4th ed. (© Garland Science 2014)

Extracellular signals can act slowly or rapidly

Contraction of muscle
Secretion of digestion enzymes

Cell growth
Cell differentiation

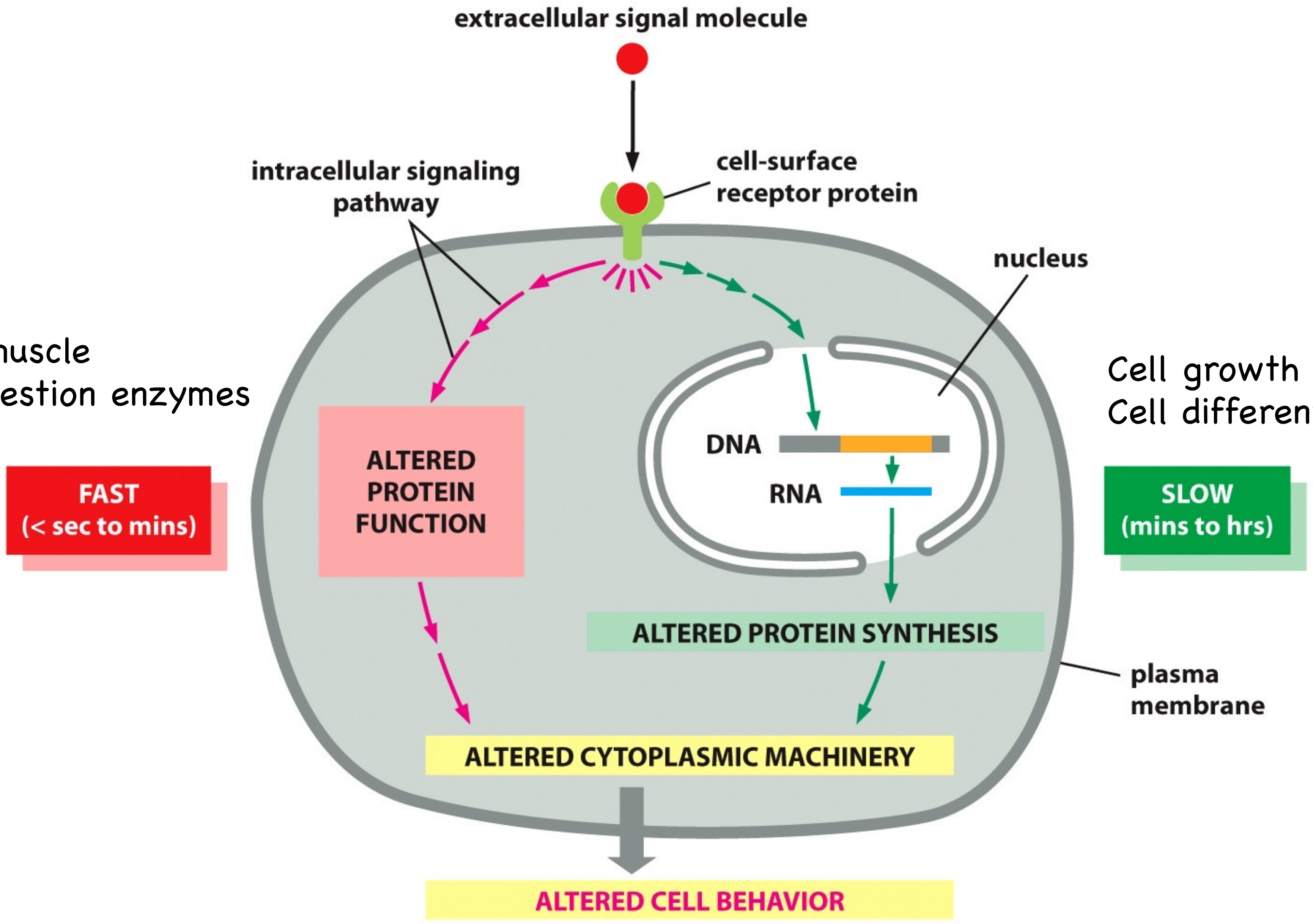


Figure 16-7 Essential Cell Biology, 4th ed. (© Garland Science 2014)

Summary I

- Cells communicate with each other via signal transduction in multiple ways

- Three phases of signal transduction
  - Reception
    - Intracellular receptors
    - Cell surface receptors
  - Transduction
    - Relay, amplify, integrate, distribute, feedback control
  - Response

How extracellular signals are received?

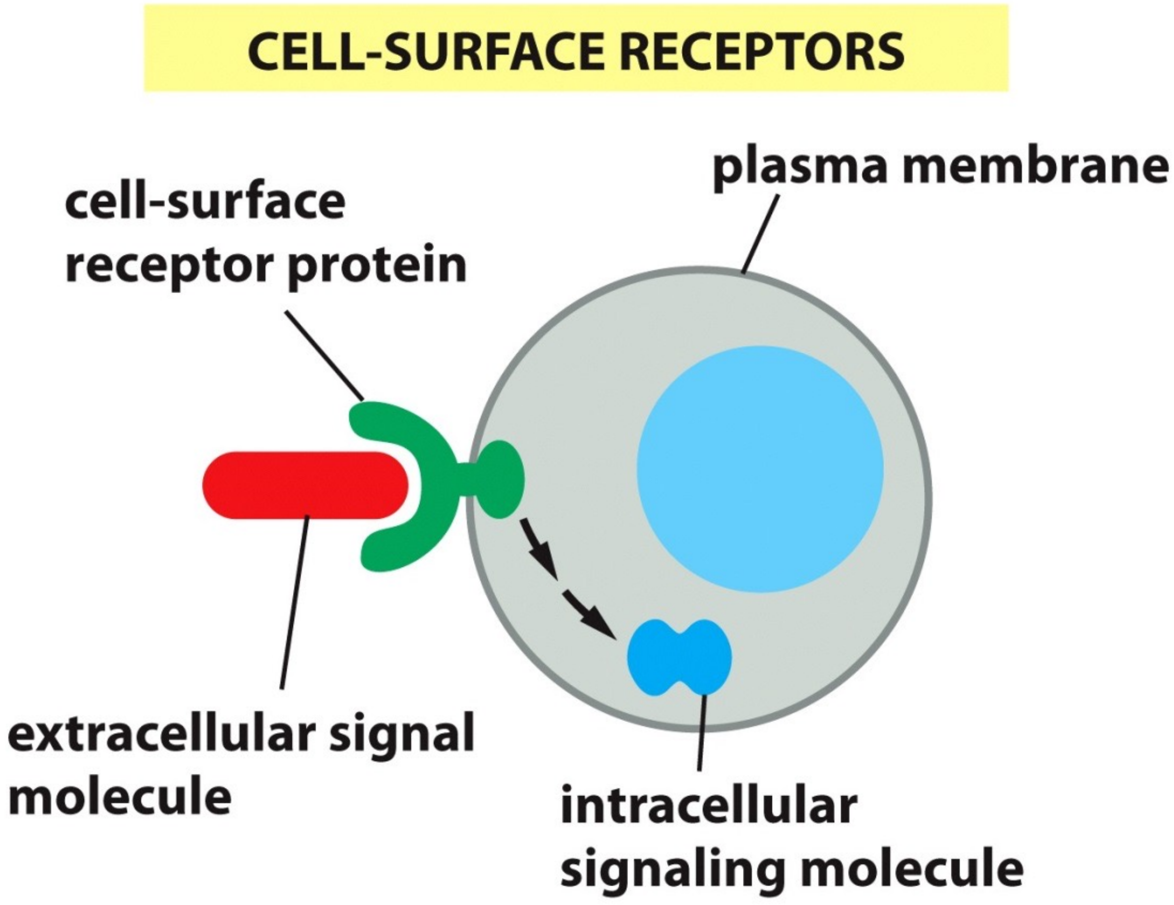


Figure 16-8a Essential Cell Biology, 4th ed. (© Garland Science 2014)

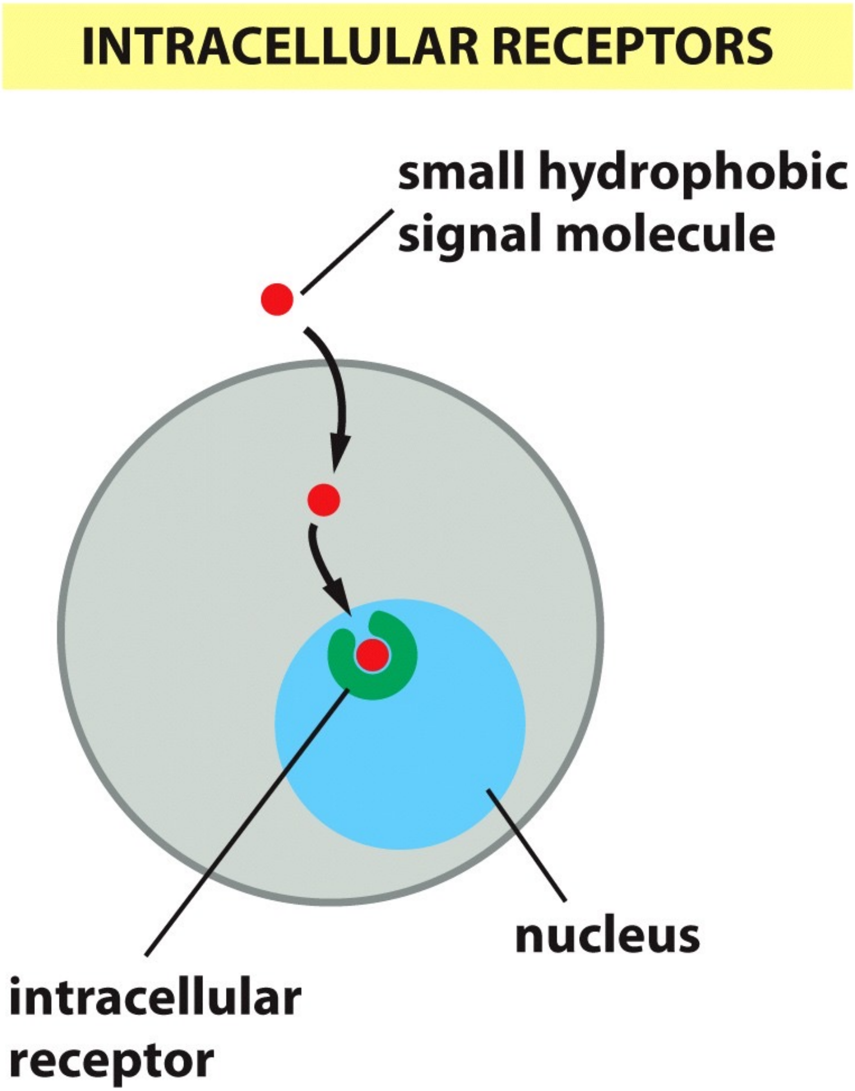


Figure 16-8b Essential Cell Biology, 4th ed. (© Garland Science 2014)

Extracellular signal molecules bind either to cell-surface receptors or to intracellular receptors.

The steroid hormone, cortisol, crosses the plasma membrane and bind to intracellular receptors to induces responses

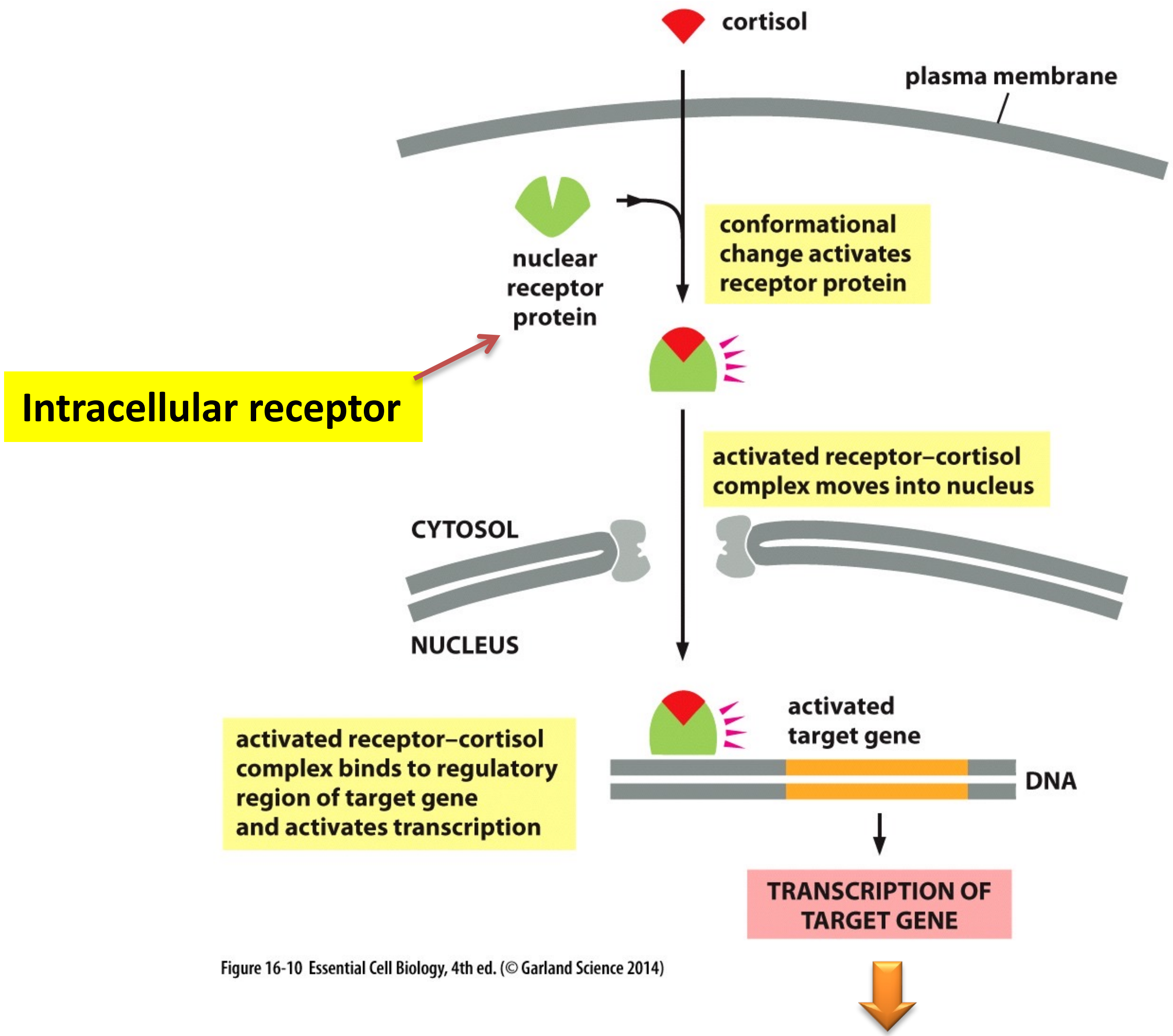


Figure 16-10 Essential Cell Biology, 4th ed. (© Garland Science 2014)

Affects metabolism of proteins, carbohydrates and lipids in most tissues

How extracellular signals are received?

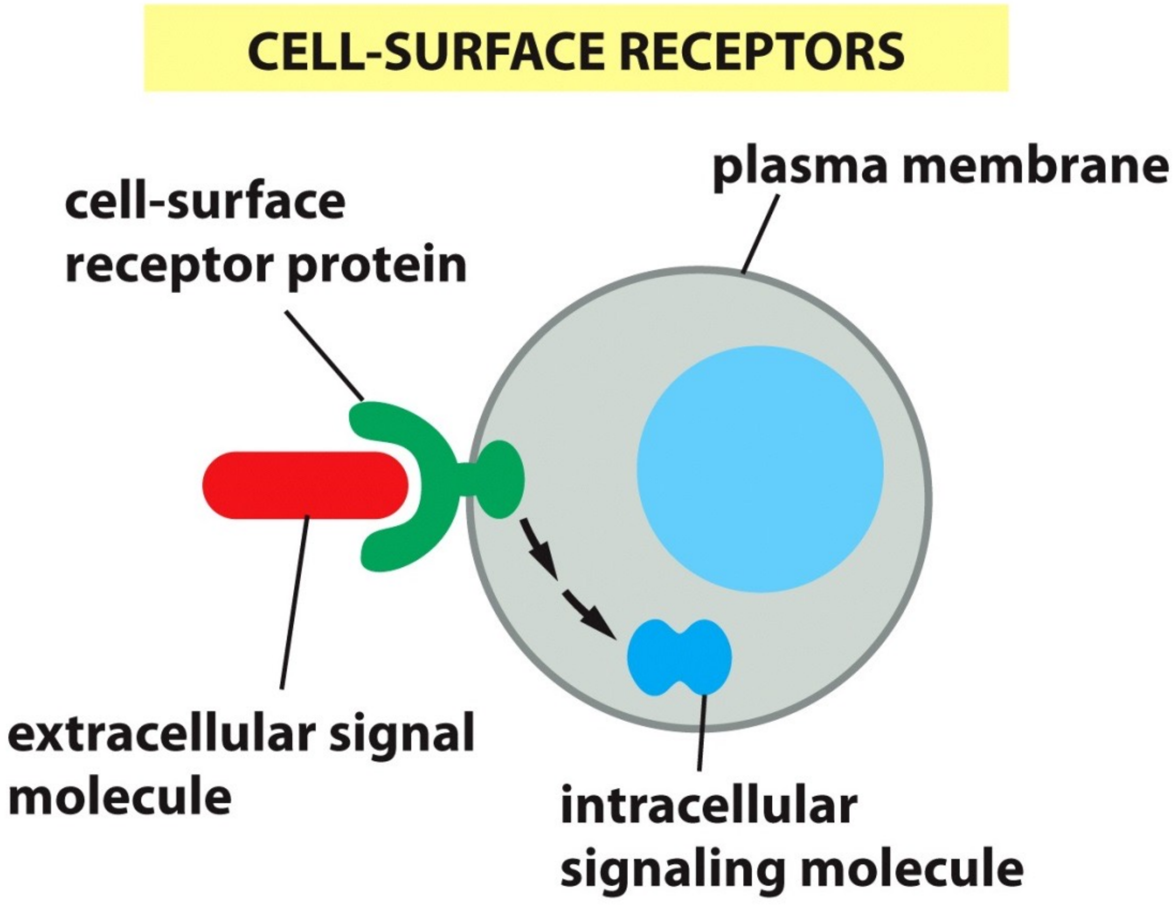


Figure 16-8a Essential Cell Biology, 4th ed. (© Garland Science 2014)

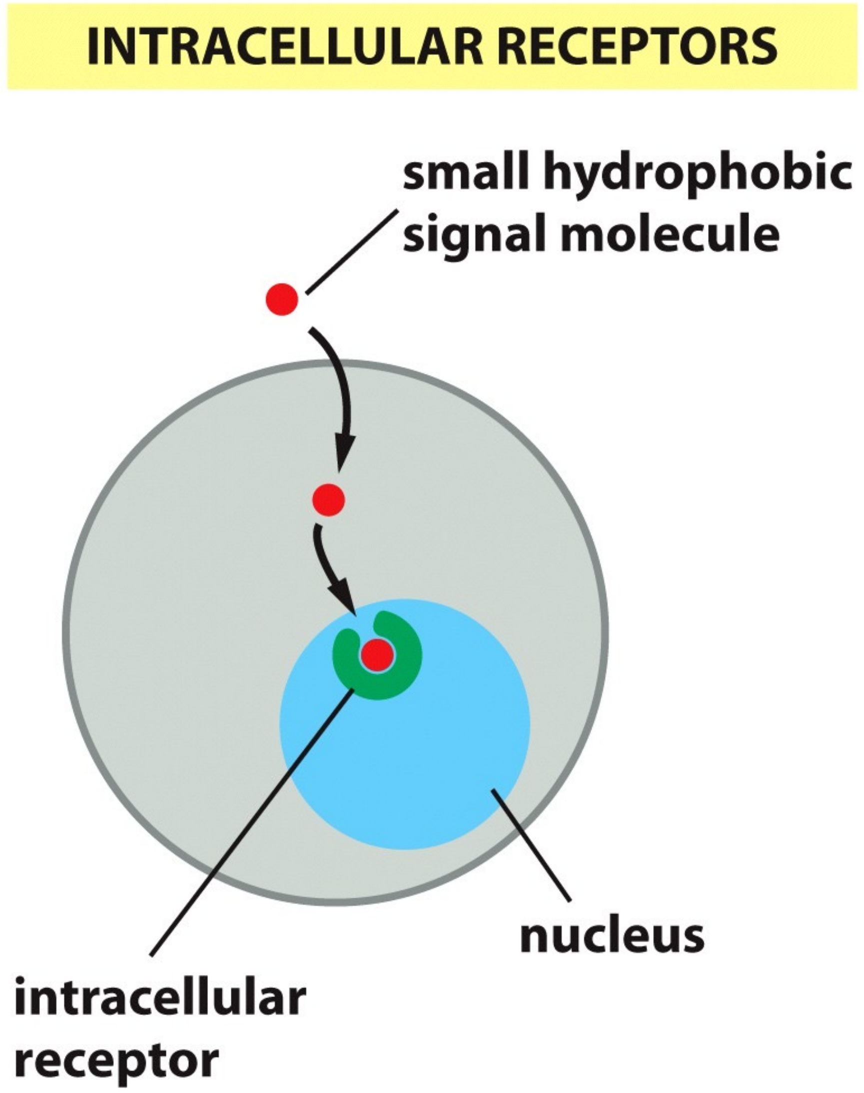


Figure 16-8b Essential Cell Biology, 4th ed. (© Garland Science 2014)

Extracellular signal molecules bind either to cell-surface receptors or to intracellular receptors.

Three main classes of cell-surface receptors

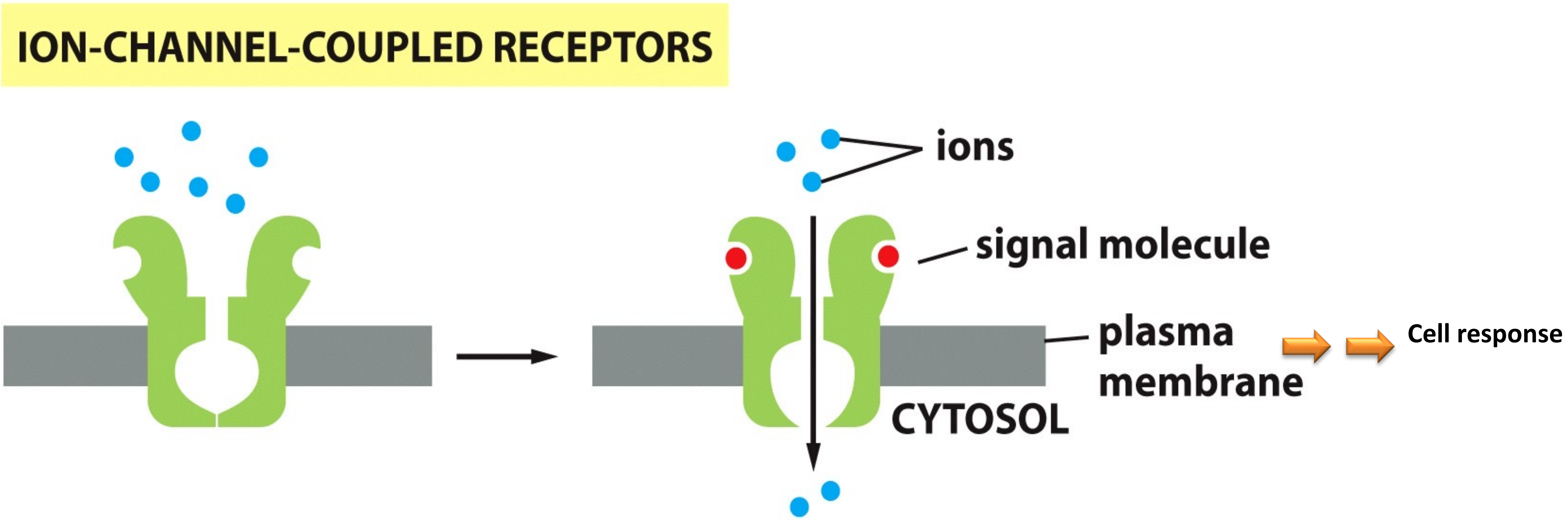


Figure 16-17a Essential Cell Biology, 4th ed. (© Garland Science 2014)

Three main classes of cell-surface receptors

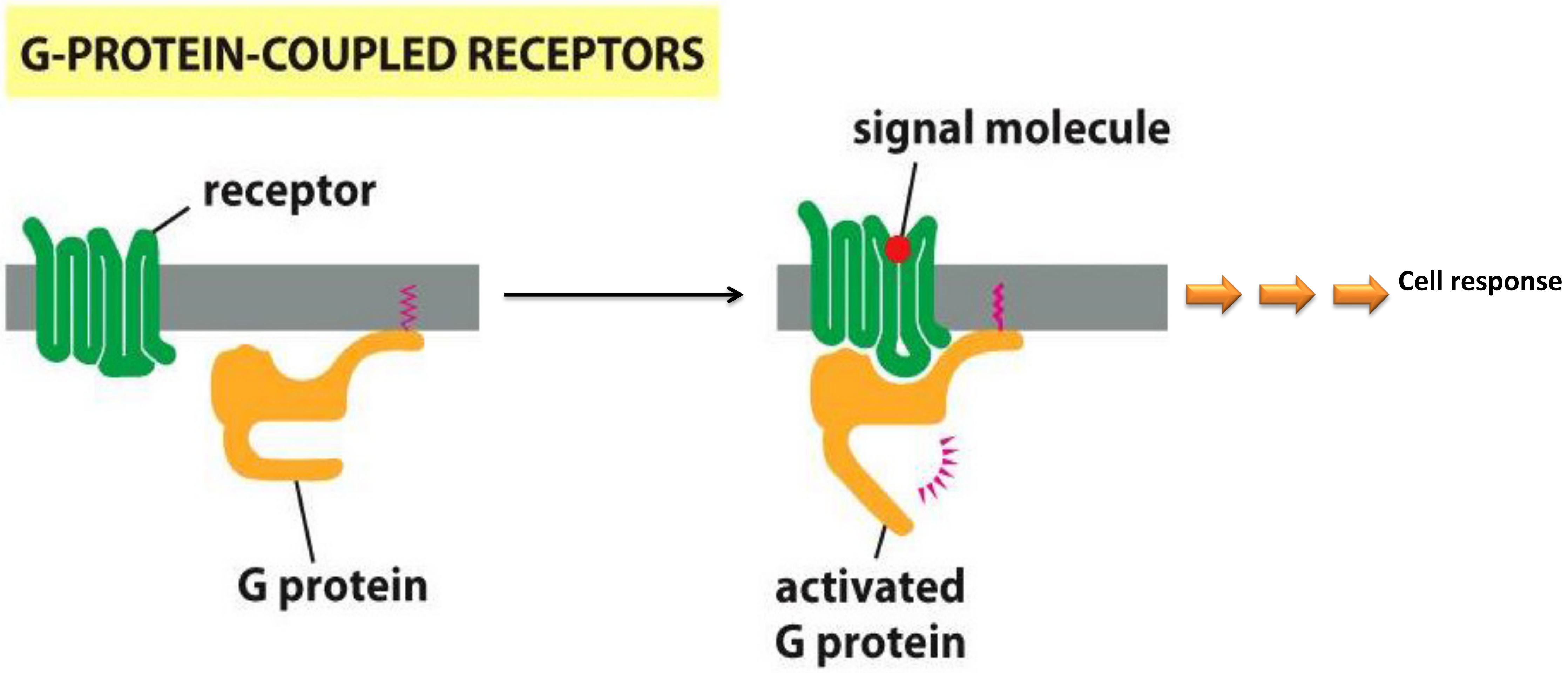
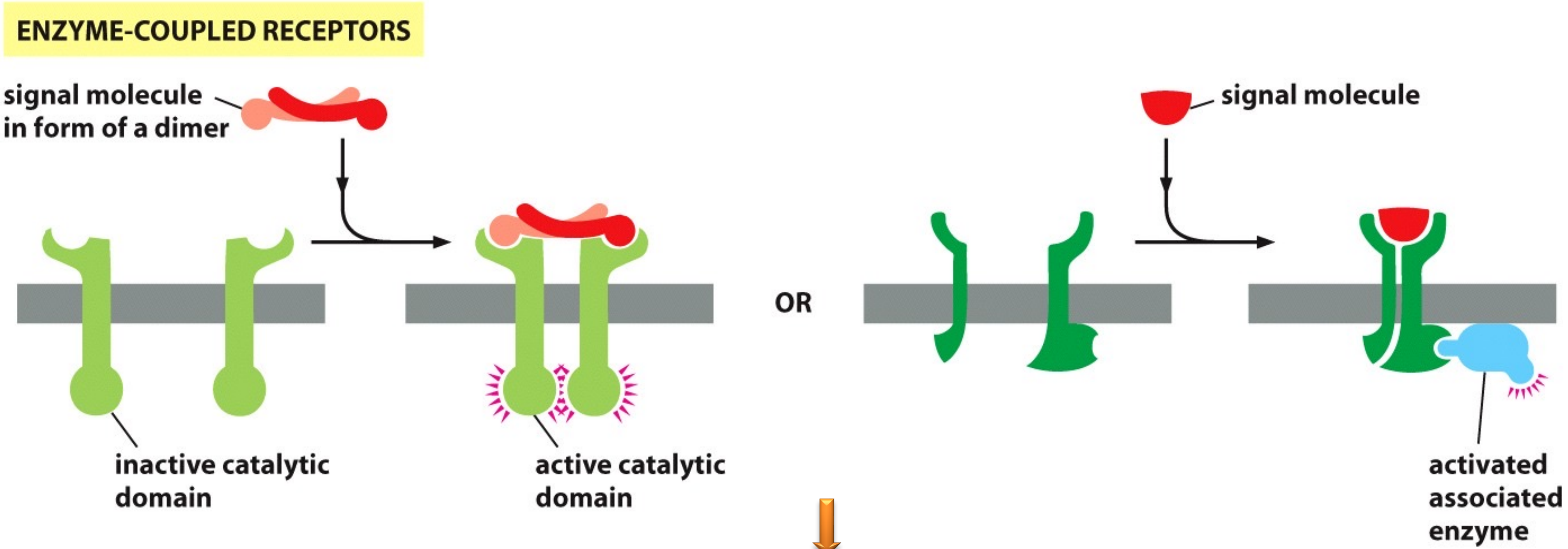


Figure 16-17b Essential Cell Biology, 4th ed. (© Garland Science 2014)

Three main classes of cell-surface receptors



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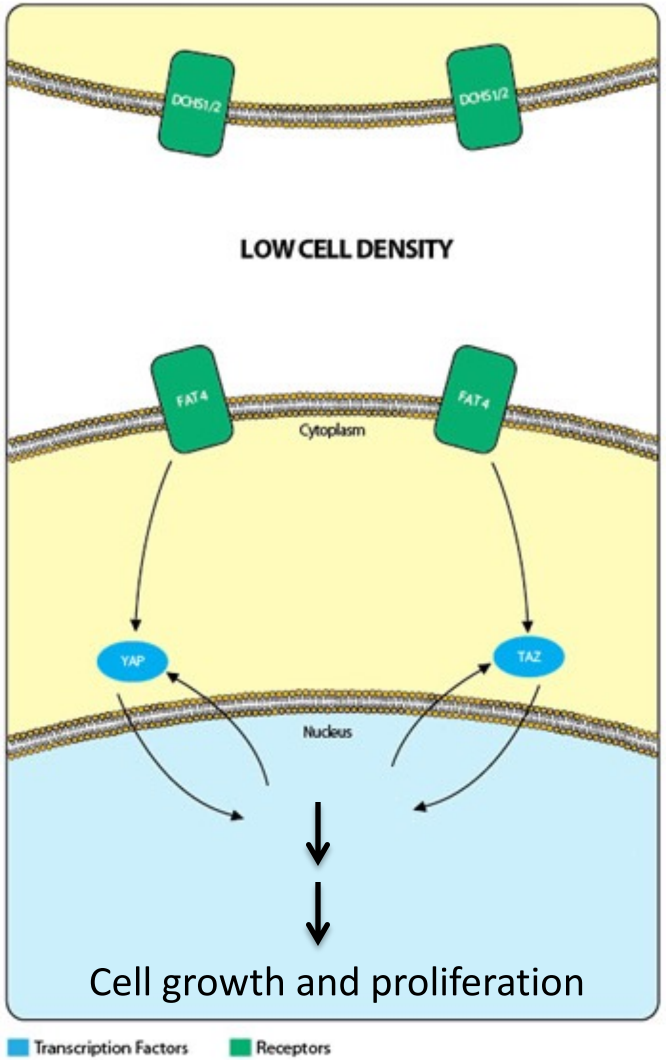
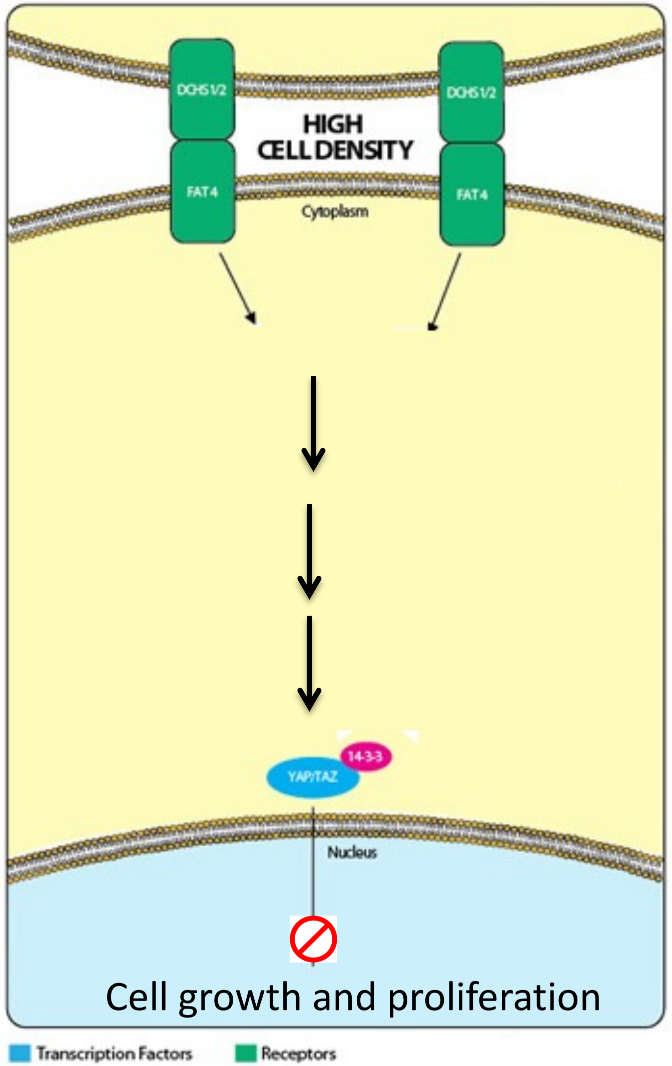
Cell response

- Ion-channel-coupled receptors
- G-protein-coupled receptors
- Enzyme-coupled receptors

The HIPPO pathway



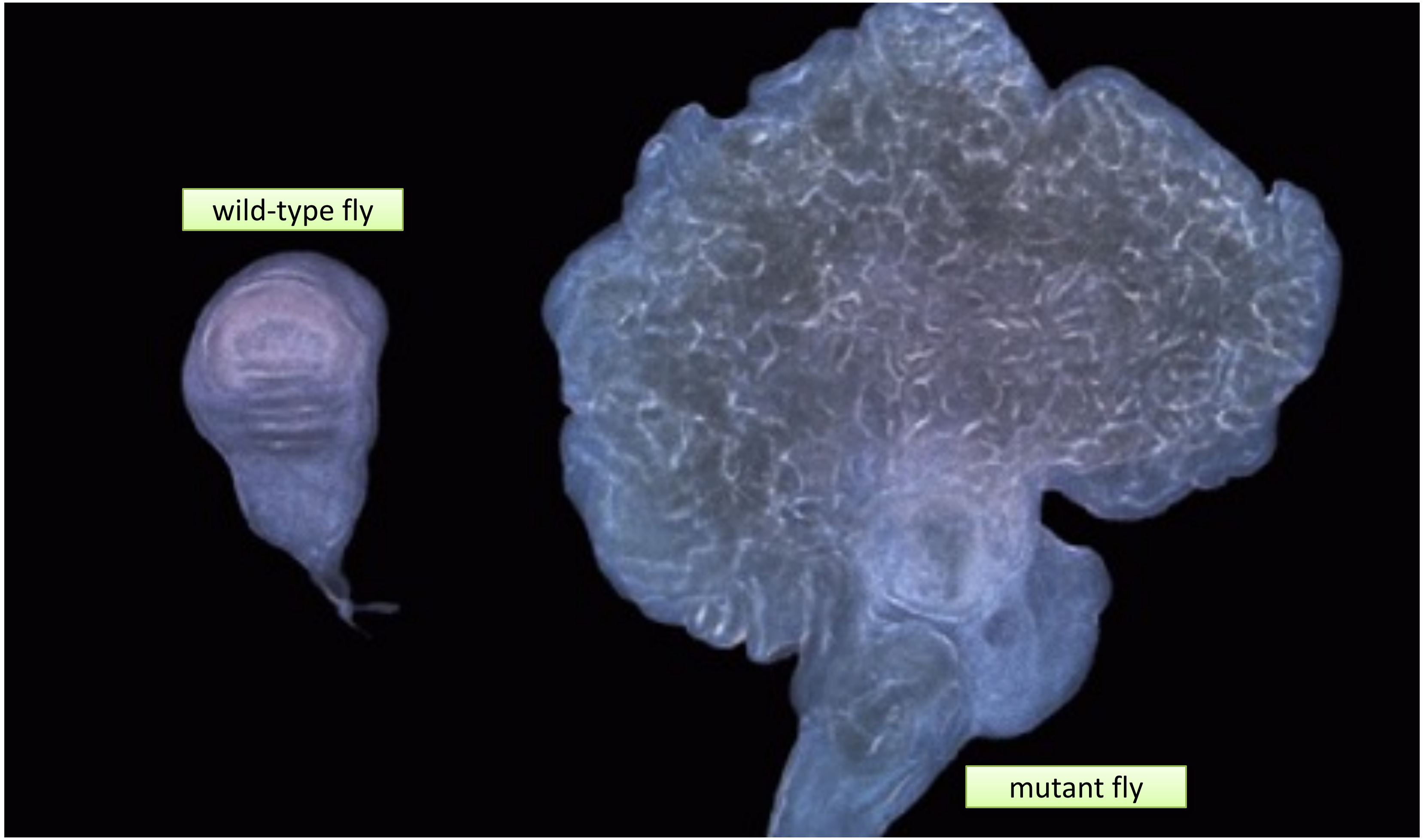
At low cell densities, the co-activators YAP/TAZ induce the expression of transcription factors that lead the pathway toward cell proliferation.



At higher cell densities, membrane-bound regulators trigger YAP/TAZ phosphorylation, thereby preventing them from entering the nucleus to induce the pro-growth transcription factors.

https://www.novusbio.com/hippo-pathway.html

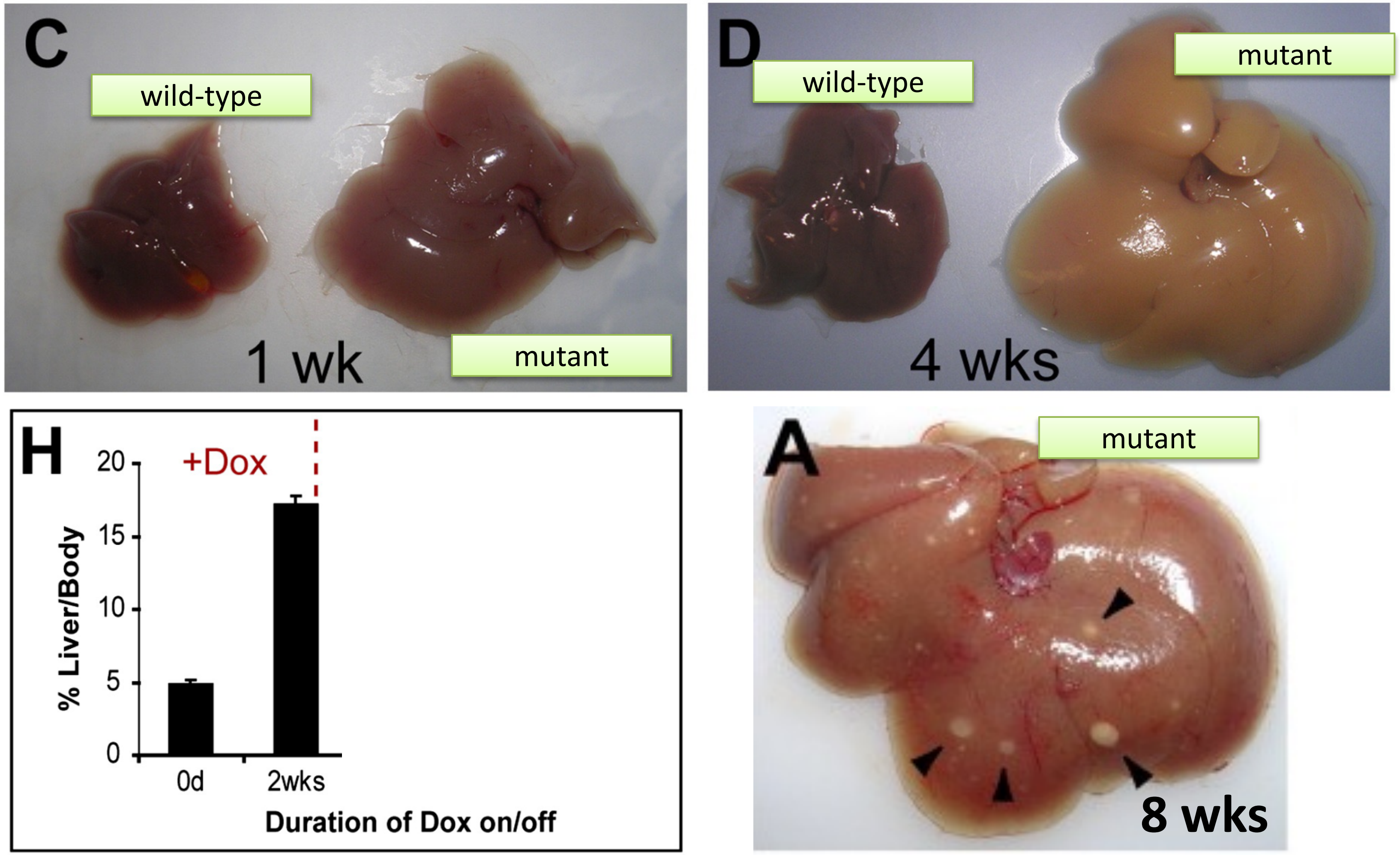
The HIPPO pathway controls organ size in fruit fly



Wing imaginal discs (a specific embryonic tissue in fly) from a wild-type fruit fly (left) and a mutant fly in which the HIPPO pathway is highly activated (right). High activation of the HIPPO pathway leads to massive tissue overgrowth.

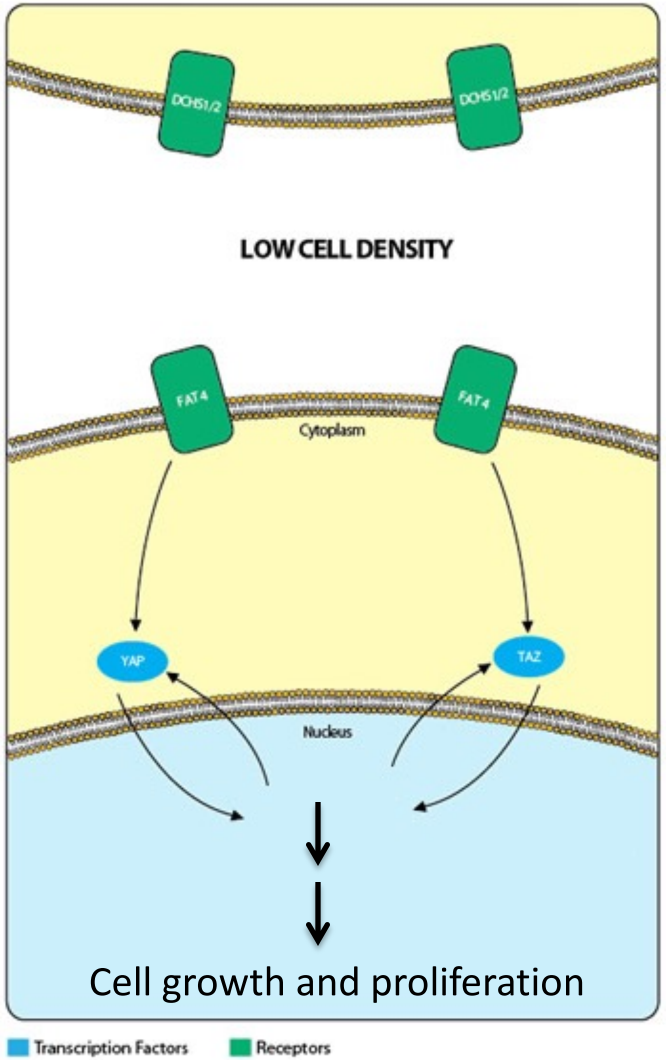
Huang, J. et al. 2005. *Cell*

Reversible Control of Organ Size by Activation/Inactivation of HIPPO pathway in mice



Jixin Dong, et al. Cell, 2007

The HIPPO pathway



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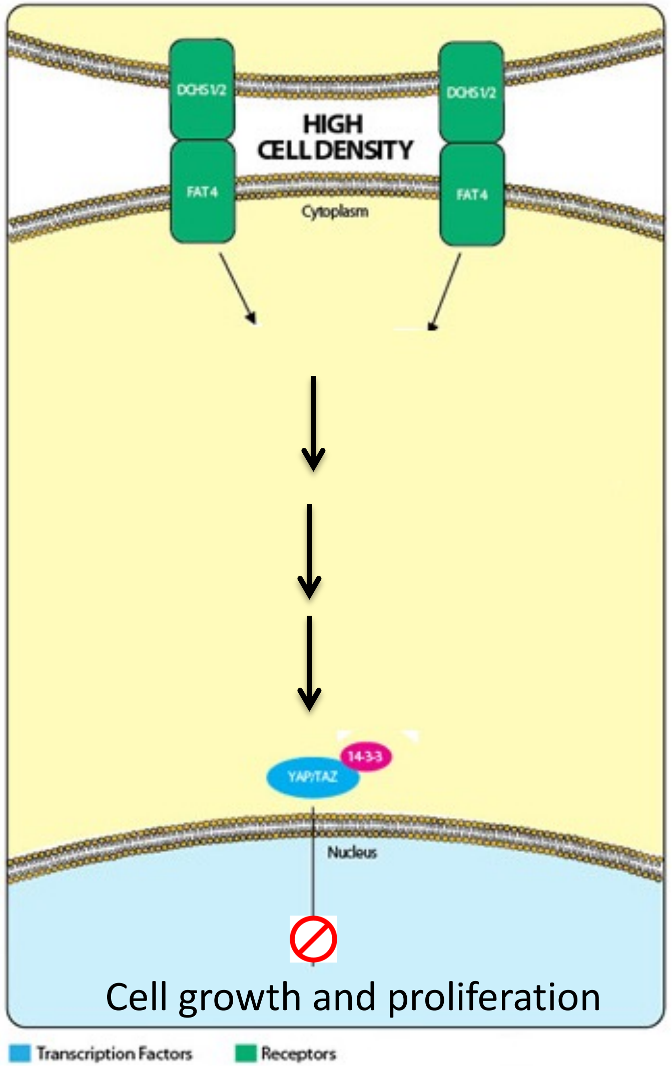

At higher cell densities, membrane-bound regulators trigger YAP/TAZ phosphorylation, thereby preventing them from entering the nucleus to induce the pro-growth transcription factors.

https://www.novusbio.com/hippo-pathway.html

Summary

- Cells communicate with each other via signal transduction in multiple ways

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